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Review

A review of literature on implementation and operational dimensions of supply chain digitalization: Framework development and future research directions

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ABSTRACT

Supply chain digitalization (SCD) significantly improves supply chain performance. This article reviews the literature on the implementation and operational dimensions of SCD. The review was conducted by analyzing the publications in SCD from 2013 to 2022. A systematic literature review of 102 publications in SCD and thematic analysis was employed in this study. The key implementation and operational dimensions, research streams, and gaps in the literature on SCD were identified. A conceptual framework for SCD is also presented. This study has attempted to summarize the literature on key implementation and operational dimensions state-of-art literature in SCD. Gaps in literature were identified and grouped categorically for better understanding and further investigations. This study could benefit industry practitioners and academicians in the logistics and supply chain community.

1. Introduction

Supply chain digitalization (SCD) has been gaining relevance in recent years. Innovative technologies and their applications in the supply chain (SC) assist in effective demand forecasting, cost optimization, and competitive advantage. Digitalization allows better connectivity among the firms, improving visibility and traceability in the SCs. This would enhance SC connectivity and add value to the network through digital integration and collaboration. The Industry 4.0 (I 4.0) revolution drives a more interconnected SC in various functional areas. Finalizing suitable strategies for the effective integration of SC to bring transparency and efficiency is one of the determining success factors. Digital transformation allows firms to gain flexibility, agility and enhance value. The end-to-end SC connectivity achieved through digitalization allows the firms to gain a competitive advantage by meeting customer demands. The benefits gained through the process of SCD involve the availability of real-time information, SC visibility, transparency, integration and collaboration, optimized intercompany logistics, and efficient inventory management (Kache & Seuring, 2017).

SCD is the adoption of digital technologies by organizations to enhance SC performance. The fourth Industrial Revolution, I.4.0, creates more interconnected functional areas across the business ecosystem, leading to value creation. The recent development in SC through digitalization has enhanced attention in practice and research. Firms adopt digital technologies like big data, radio frequency identification, cloud computing, Internet of things (IoT), and artificial intelligence to respond effectively to dynamic business environments. Applying emerging digital technologies makes SCs more proactive and competitive in the market (Buyukozkan & Gocer, 2018).

A literature review conducted on peer-reviewed articles published from 2000 to 2017 shows that the main drivers of the digital supply chain (DSC) include technologies, integration, digitization, collaboration, and coordination. There is a requirement to investigate further the DSC development and its conceptualization (Iddris, 2018). I 4.0 is evolving by introducing digital technologies encouraging SC partners' value chain and connectivity. It enhances operational efficiency and the introduction of new products, services, and business models (Luz Martin-Pena et al., 2018).

Liao et al. (2017) conducted a literature review of progress in industry 4.0 by considering articles published online until June 2016 and presented specific data analysis on four key concepts. Garay-Rondero et al. (2020) developed a referential and systematic model considering the concepts of SCM and digital technologies and its trends toward SCD. Zekhnini et al. (2020) proposed a conceptual model defining the critical components that mold DSCs through Industry 4.0 based on the literature from 1989 to 2019.

Through practical analysis of the literature published from 2013 to 2022, this study provides a comprehensive view and guidance for future research directions in SCD. Various theories considered in information sharing in SC research are organization information process-

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ing theory, contingency theory, cost economics, resource dependency, resource-based view, and relational governance theories. However, SCD can be underpinned effectively by the Organizational Information Processing Theory (OIPT). Accordingly, this review article advances the understanding of the process of SCD, considering various aspects of OIPT. The concept of OIPT provides better insights into the application and utilization of digital technologies for enhancing SC performance in a dynamic business environment. Kamble et al. (2020) have highlighted the relationship between SC performance and information processing capability based on the data analytics concept.

Literature has mentioned the adoption and development of information systems in organizations to enhance processing capability. The contingency theory and dynamic capability view theory also add to the concept of OIPT. Contingency theory addresses the requirement for generating strategic fit among information processing demand and capacity. The dynamic capability view considers the application and use of digital technologies to generate better organizational capabilities, including effective collaboration and performance. A review of the existing literature suggests understanding various dimensions based on theoretical underpinning, which will contribute to the literature and advance further research in SCD.

The vast advancement in the technology and its applications demands an exploration to identify the theoretical gaps and its conceptualization, thus contributing to the literature on SCD. As the research in SC is progressing rapidly, a thorough review of the recent literature by focusing on the implementation and operational dimensions of SCD is a timely study to bring better insights and clarity on the status of research in the SCD area. There is a gap in the existing literature on SCD to understand the implementation and operational dimensions and emerging areas that demand further investigation.

This study is unique as it integrates various implementation and operational dimensions of SCD, research streams, gaps in the literature, and future research areas in SCD. A systematic review of the literature is done to identify gaps in the literature on SCD. Even though conceptual frameworks are available for the DSC, one that considers the implementation and operational dimensions of SCD are limited. In this study, we have elaborated upon the main research themes of SCD and have proposed a new conceptual framework. The conceptual framework and branching of its structure based on implementation and operational dimensions of SCD facilitate consideration of various elements influencing the digitalization process.

Studies on operational feasibility, adoption, and maturity in digital technologies are scarce. Nevertheless, the gaps in the literature in these areas provide a compelling justification to proceed with further study on the process of SCD. Thus, the results of our study can contribute to the existing literature and assist practicing managers in decision-making for effective SCD. It also opens up new research areas for further investigation in the digital transformation of SC in various industries.

The main objectives of this research are as follows:

(i) Analyzing the evolution and extent of work done in the area of SCD on implementation and operational dimensions based on search in scientific databases,
(ii) Classifying and grouping major and sub-major elements of implementation and operational dimensions in the area of SCD,
(iii) Identify and uncover the emerging trends and topics in the area of SCD, affecting implementation and operational dimensions of SCD and
(iv) To enable the researchers to identify the research gaps and provide future research directions.

A systematic literature review (SLR) identifies and classifies literature based on key factors and significant strategies. Hanafizadeh and Zareravasan (2020) opine that SLR assists in suggesting future research directions and lead to better insights on a topic. SLR also facilitates evidence-based decision-making. This study has focused on SCD by adopting an appropriate methodology for searching, selecting papers, analyzing, and synthesizing logical and defendable. This review has chosen articles from reputed peer-reviewed journals by following well-defined inclusion and exclusion criteria. SLR is a robust and auditable method that assists in developing theories and arriving at conclusions (Wilding & Wanger, 2014).

Further, conducting a SLR helps to identify and explore the focused area in-depth. Literature on SCD published in online scientific databases from 2013 to 2022 was checked to achieve our research objectives. The methodology adopted assists in identifying influential studies, prospective research areas, and proper classification of the topics related to SCD.

The major contributions of this research in the area of supply chain digitalization are as follows:

- Major elements of supply chain digitalization from the implementation and operations perspectives are analyzed and presented.
- Emerging trends, gaps and future research areas in SCD are identified.
- Contributes to the literature on SCD, which has wide theoretical and practical implications.

This article is further organized as follows. Section 2 deals with the existing literature review in the area of SCD. The review methodology used in this study is detailed in Section 3. A detailed discussion on developing a conceptual framework for the dimensions and components of SCD is presented in Section 4. Emerging research streams on SCD and gaps in the literature are given in Section 5. The discussions, synthesis of the literature, practical implications, future research directions, and limitations of the research are presented in Section 6.

2. Existing literature – review

This section summarizes the earlier reviews and the requirement for a state-of-art literature review focusing on implementation and operational dimensions in SCD. Literature on SCD published in recent years is increasing due to the topic's relevance and enhanced interest in the subject. We justify the need to do a SLR and develop a framework by reviewing and summarizing the literature on SCD. The existing literature indicates that the number of publications on SCD has increased during the last few years, which denotes the significance and interest among practitioners and academicians in the SC community. Hence, this research will facilitate in identification of new research streams, gaps in the literature, and topics that need further investigation. This section also highlights the requirement of shifting towards digitalized SC from traditional SC and its effective management.

2.1. Digital supply chain

Literature reveals that past researchers have examined various aspects of digitalized SC as the relationship between IT and SC performance (Samadi & Kassou, 2016); intelligent manufacturing in the context of I 4.0 (Zhong et al., 2017); development of DSC leading to the high-level conceptualization and theoretical development (Iddris,2018); key prospects and limitations of DSC (Buyukozkan & Gocer, 2018); main components of I 4.0 and impact on DSC (Garay-Rontero et al., 2020); trends in the digitization of SC (Marmolejo-Saucedo & Hartmann, 2020) and firm's orientation in adopting I 4.0 and DSC (Gupta et al., 2020). Novak et al. (2021) analyzed the decision making information systems through real-time sensor networks and big data analytics through AI in sustainable I 4.0.

2.2. Supply chain in industry 4.0 era

I 4.0 is evolving by introducing digital technologies encouraging SC partners' value chain and connectivity. Hofmann and Rusch (2017) have proposed the concept pillars based on I 4.0 definitions: (i) cyber-physical systems, (ii) IoT, (iii) internet of services, and (iv) smart factory. Key ingredients to assess an organization's readiness for I 4.0 are the digitization of SC, organizational strategies, top management involvement and

Table 1
Summary of relevant studies in the area of SCD.

Author(s)	Contribution to literature
Liao et al. (2017)	Conducted a literature review of progress in industry 4.0, considering articles until June 2016 and presented specific data analysis on four key concepts.
Iddris (2018)	Literature review from 2000 to 2017. Highlighted the need to investigate DSC development and its conceptualization
Buyukozan and Gocer (2018)	Review of DSC, its importance, and its applications.
Luz Martin-Pena et al. (2018)	I 4.0 enhances operational efficiency; Introducing new products, services, and business models needs a better understanding of SC.
Garay-Rondero et al. (2019)	They developed a referential and systematic model considering the concepts of SCM and digital technologies and its trends towards SCD.
Zekhnini et al. (2020)	They proposed a conceptual model defining the critical components that mold DSCs through Industry 4.0 based on the literature from 1989 to 2019.
Han et al. (2021)	They identified and focused on the challenges and solutions in digitizing supply chains, which revealed that developed solutions could positively and significantly impact SCD.
Alsufyani & Gill, 2022	They studied the impact of digitalization on organization performance outcomes.

commitment. I 4.0 enhances operational efficiency (Luz Martin-Pena et al., 2018). Tortorella et al. (2020) addressed the mediating role of organizational learning capabilities at different contextualizations between I 4.0 technologies and operational performance.

Seyedghorban et al. (2020) have adopted a mixture of quantitative and qualitative review methods to identify the related works in the area of SCD. Wamba and Queiroz (2020) have done a SLR of 149 peer-reviewed articles related to case studies in SC from 2010 to 2017. The contribution of digitalization to the manufacturing sector finds emphasis in the literature. Zangiacomi et al. (2020) have also investigated the managerial practices for adopting I 4.0 technologies. Kittipanya-Ngam (2020) has developed a framework for digitalizing food SC through case studies based on three leading companies and highlighted the practices, opportunities, and challenges faced by the food manufacturers in Thailand. Bordeleau (2020) highlighted the significance of business intelligence and analytics in value creation. Boehmer et al. (2020) studied the potential of IoT technology and its application in developing new production businesses and operating models.

2.3. Supply chain digitalization techniques

Di Vaio and Varriale (2020) have highlighted the relevance of rethinking and redesigning data management by adopting digital platforms for port operations in the sea-land SC. Ukko et al. (2020) explored the relational mechanisms and performance measurement in digital service SC that facilitate closer collaboration in performance measurement. The impact on knowledge management of small and medium enterprises through knowledge sharing and knowledge transfer in the context of green SCM and the moderating effect of collaborative communication was studied (Pan et al., 2020a). Asthana et al. (2020) have analyzed the performance measurement of third party logistics sector by focusing on relationship management and its improvements in the competitiveness of logistics service providers.

The control of an individual product's life cycle from design to production through the effective engagement of firms in all the activities can be facilitated through digital encapsulation (Engel et al.,2017). Frank et al. (2019); Dolgui et al. (2019b), Ivanov et al. (2019), and Zennaro et al. (2019) have introduced various engineering and management frameworks of I 4.0, highlighting the customizable supply and manufacturing networks. Furthermore, the existing studies pointed toward forming cyber-physical SCs (Ivanov et al., 2019; Panetto et al., 2019). The studies on the application of data analytics and its use to enhance demand forecasting, production flexibility, SC visibility and transparency, and its contribution to effective SCM are also found in the literature.

A state-of-art SLR on SCD considering the implementation and operational dimensions of SCD and identifying the dominant research streams and gaps in the literature is minimal. Most articles focus on the DSC and its significant drivers and technologies. Even though the articles published on DSC have highlighted the requirement for more investigation in certain areas, only a few researchers have investigated the key dimensions, benefits, and challenges of SCD by identifying the dominant research streams and gaps in the literature that enhance the knowledge base. A summary of the relevant studies in the area of supply chain digitalization is provided in Table 1. The need to investigate further the development of DSC, its conceptualization, significance and applications, the evolvement of business models, trends towards SCD and the outcome of SCD on an organization's performance were found in the literature.

Hence, this study is timely on SCD. The review methodology applied in this study is discussed in the next section.

3. Review methodology

Studies reveal that many papers have been published in the area of SCD. Also, some literature review papers on SCD are found. However, it is noted that there is limited research in analysing the implementation and operational aspects of SCD relating the theories and interlinking with various elements leading to SCD. Also, only a few conceptual frameworks for SCD have been reported. Accordingly, this research identified and elaborated on five major concepts, mentioning various aspects and dimensions involved in each theme. A conceptual framework is developed underpinning theoretical elements, themes and dimensions, thus identifying new research streams and research gaps on SCD.

This research has followed a systematic literature review method considering the principle of the deductive research approach, similar to the methodology adopted by Frederico et al. (2019). Further, a qualitative literature assessment was applied to the SLR to determine the relationship between SCD and various aspects. The SLR process consists of three phases: planning, screening and reporting. Further, a concept-oriented method was used to identify the significant themes that assist in developing a conceptual framework for SCD.

The research questions were addressed by carrying out a SLR. Wilding and Wanger (2014) have stated that SLR is an auditable and robust method that facilitates theories and research conclusions. Therefore, SLR was conducted to address the following four research questions (RQs).

1. How can the process of implementation and operational dimensions of SCD be evaluated and understood? (RQ1)
2. What elements constitute the implementation and operational dimensions of the SCD process? (RQ2)
3. What are the emerging trends and topics in the area of SCD? (RQ3)
4. What are the critical research gaps and future research directions related to SCD? (RQ4)

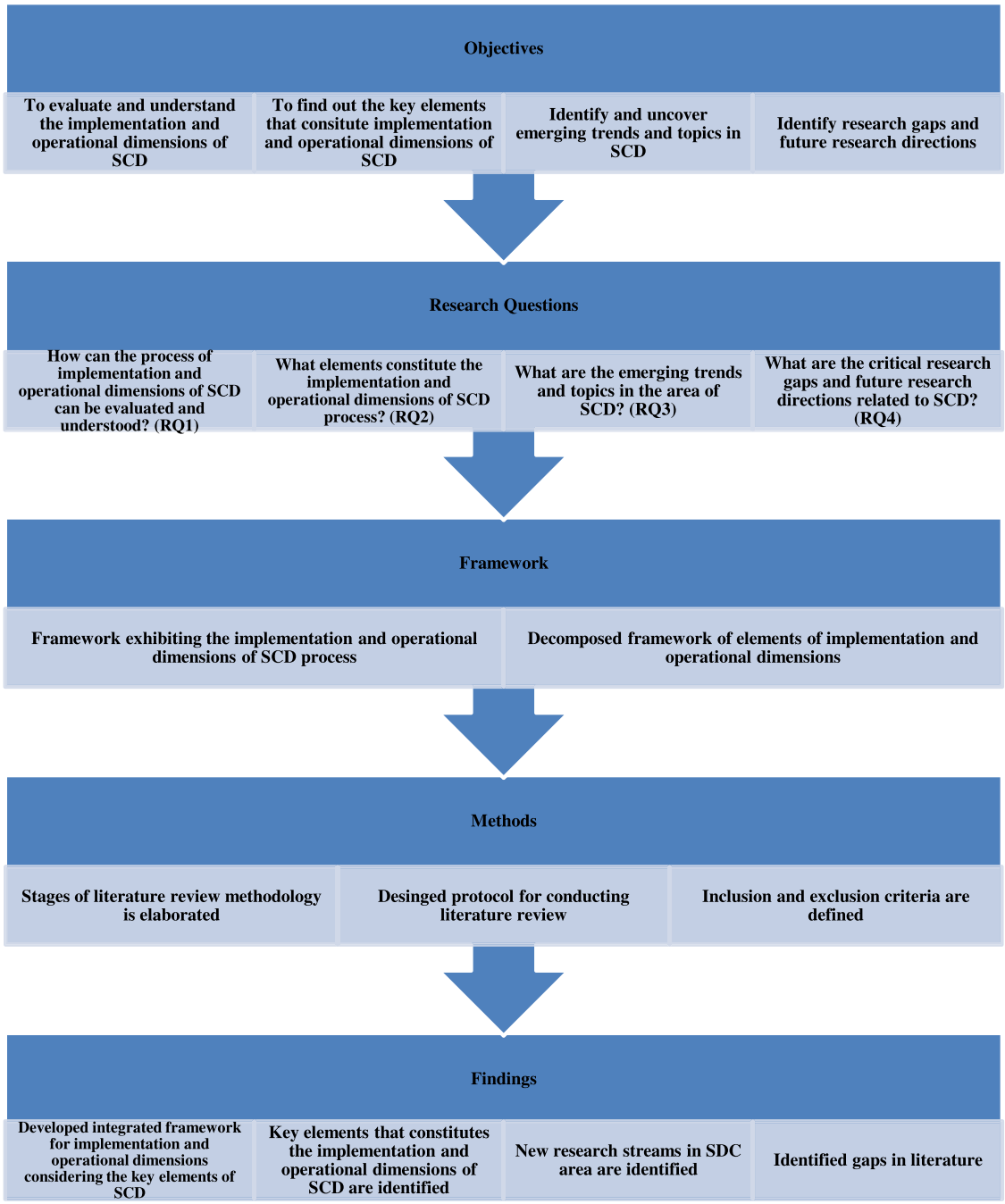

Fig. 1. Research process.

A diagrammatic representation of the research process linking the research questions with research objectives, framework, methods, and findings is shown in Fig. 1.

In the initial phase, the research objectives were finalized and then followed by developing the research questions. We further elaborate by mentioning the stages and protocol for conducting the review and finalizing the inclusion and exclusion criteria. Subsequently, an integrated framework for implementation and operational dimensions considering the key elements of SCD is developed. This has resulted in identifying new research streams and gaps in the area of SCD.

3.1. Data collection

SLR was conducted in three phases, as proposed by Tranfield et al. (2003). It includes planning, conducting, and reporting. The planning phase involves identifying the search databases and criteria for the search. Then, the search is carried out, followed by a screening of search results, extraction, data analysis, and synthesis. The outcome of the investigation is presented in the reporting phase in a structured manner.

This research followed a predefined search protocol to review all relevant articles on the subject comprehensively. In the first step of identification, peer-reviewed articles in SCD are identified based on the shortlisted keywords search. A search using keywords such as “supply chain digitalization” and “digital supply chain”, “digital transformation”, “Industry 4.0″, “Supply Chain 4.0″, “Logistics 4.0″, “Blockchain,” “cloud computing,” “industry 4.0″, “artificial intelligence”, “big data analytics”, “IoT”, “machine learning”, “drones”, “3D printing”, “cyber-physical systems”, “additive manufacturing”, “virtual and augmented

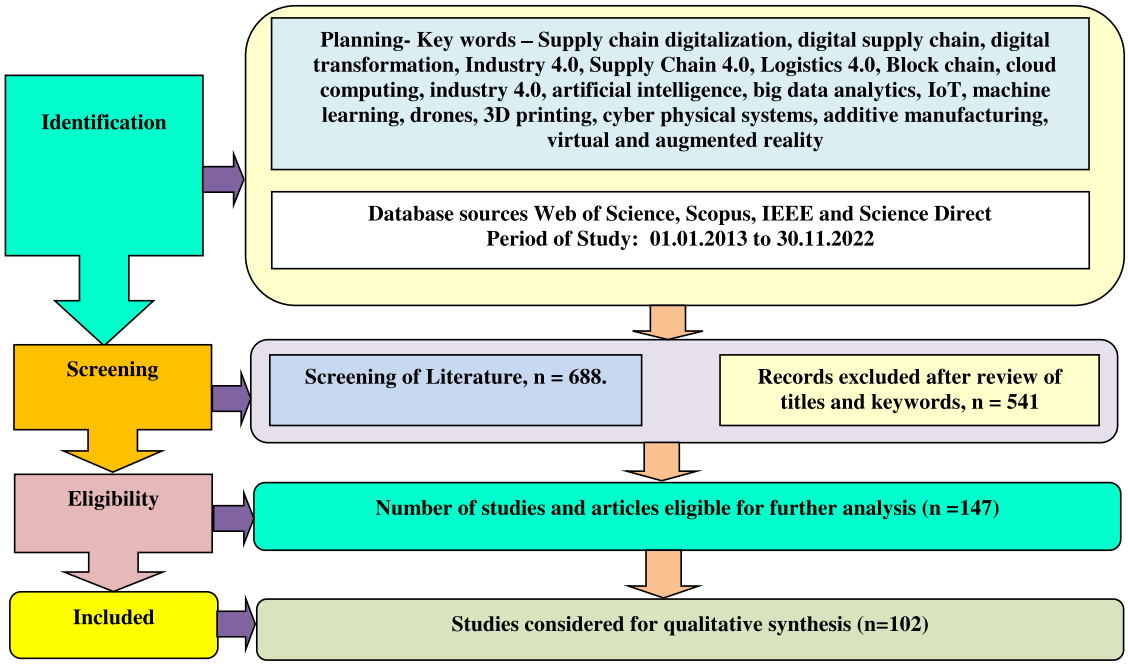

Fig. 2. Systematic literature review- Protocol.

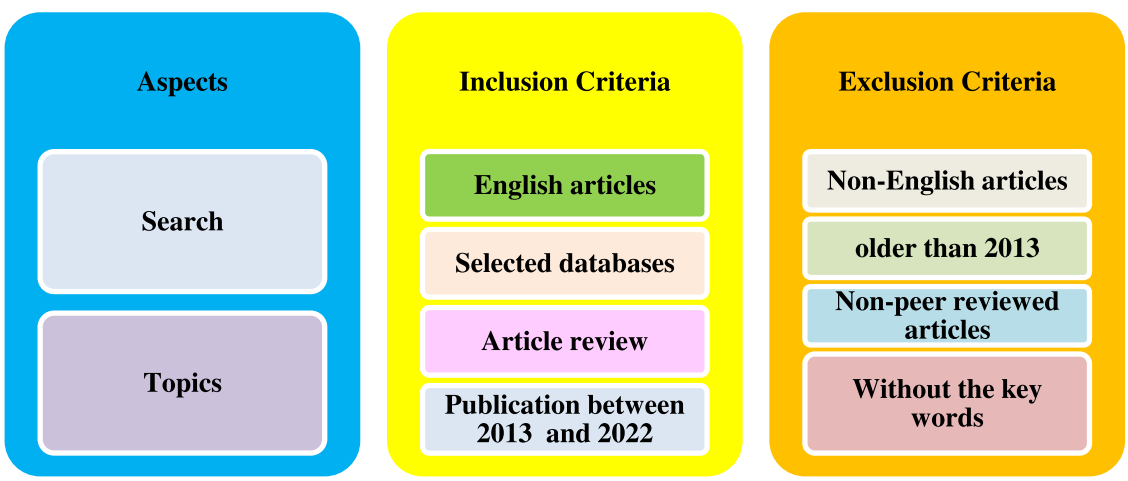

Fig. 3. Inclusion and exclusion criteria.

reality" in the databases of Google Scholar, IEEE Explore, Web of Science and Scopus returned many research papers. The four main databases identified have maximum coverage of the most quality journals related to the subject under review in this research. The results of this study covering such databases also provide easy access to articles by the researchers.

This was further shortlisted to 102 articles spread across 63 journals based on the study's relevance to SCD. The SLR protocol followed in this research with the inclusion and exclusion criteria followed are shown in Fig. 2 and Fig. 3, respectively.

The search was conducted on databases like Web of Science, Scopus, IEEE and Sciencedirect. Only the publications appearing in English from 1st January 2013 to 30th November 2022 were considered in this study. Accordingly, 688 articles were shortlisted. In the next step, the shortlisted articles were screened, and 541 were excluded based on a review of titles and keywords, resulting in shortlisting of 147 of them. In the final step, these 147 articles were reviewed thoroughly. Considering the context of implementation and operational dimensions of SCD, only 102 articles were finally considered for analysis in this research.

Exclusion criteria- Non –English Articles, non-peer-reviewed articles older than 2013, without the keywords.

3.2. Data analysis

A format was designed in Microsoft Excel to compile the shortlisted articles. The data extraction and analysis for this research were based on the research questions. The format devised mainly covered aspects like demographics, including authors' details, the paper's title and the year of publication. The objectives of the articles and major contributions to the literature were also compiled. The format was verified and rigorously tested on a few randomly selected papers. Hence, the data considered

in this research and the format designed for data analysis ensure the validity and authenticity of the data collection and its analysis.

4. Development of conceptual framework for supply chain digitalization

This section deals with discussions and technologies for SCD, thematic analysis and development of conceptual SCD framework focusing on implementation and operational dimensions of SCD.

4.1. Supply chain digitalization

Digitalization of SC allows the integration of data and information by assisting various functions of SC processes (Mussomeli et al., 2016). Digital technologies help in the real-time transmission of information and support knowledge management practices. Advanced digital technologies allow real-time information transmission and support knowledge management practices (Wilkesmann and Wilkesmann, 2018). The desire to adopt new technologies will transform SC (Xue et al., 2013). Advanced technologies give companies a competitive advantage through higher revenue and value addition (Buyukozkan and Gocer, 2018). The digitalization process development will enable organizations to manage their SC activities remotely (Lyall et al., 2018). Based on the articles shortlisted for review, it is noted that there is a lack of knowledge in implementing and effective utilization of digital technologies.

The details of 102 articles considered for literature review in SCD are given in Appendix A. The research article's reference, the aim of the study, and implications and contributions are detailed in the Appendix. It was found that the 102 articles shortlisted were spread across 63 journals. The diagrammatic representation of the distribution of journal articles shortlisted across various journals is illustrated in Fig. 4.

Supply chain integration (SCI) through digitalization involves how a company adopts digital technologies in conducting its day-to-day transactions. It aims at streamlining the flow of products, information and funds from suppliers to customers, thereby ensuring efficiency and accuracy in SC processes. In today's globalized and highly competitive world, SC strategies and objectives must be aligned with the business objectives to achieve maximum benefits. It adds value to the customer by gaining customer satisfaction and market share. The three main pillars of the SCI framework are information integration, coordination and resource sharing and organizational relationship linkage.

The key terminologies of SCD found in the articles were analyzed. Accordingly, eighteen prominent terminologies appearing in the review articles were shortlisted. A diagrammatic representation of the number of occurrences of the key terminologies is shown in Fig. 5. It is found that digitalization, Industry 4.0, digital manufacturing and digital transformation are the most common terminologies, showing the SCD process's significance. The rest of the terminologies related to SCD like digital supply chain, smart factory, block chain, smart supply chain, cloud manufacturing, data analytics, information and communication technology, DSC framework, internet of things, supply chain 4.0, cyber-physical systems, cloud manufacturing, technology enablers, process automation and supply chain transformation have also found to be evolved during the digital era, which signifies the role of digitalization in the supply chain.

Further, the year-wise distribution of publications from 2013 to 2022 is illustrated in Fig. 6. Among the articles shortlisted for review in this study, thirty-one publications were from 2019, followed by thirty in 2020. In recent years, the higher number of publications amidst pandemic situations shows the necessity of digitalization and digital technologies in the SC. The trend in the year-wise distribution of articles envisages the requirement of a study considering the relevant publications till date to find the literature gaps and identify new research streams. An analysis of SCD aspects also facilitates identifying the key areas addressed in the literature.

4.2. Components and technologies for SCD

Research in information systems has continued to evolve consistently in recent decades on aspects like intellectual core, diversity, and the impact of information systems. As a result, it has advanced considerably over time while serving its core identity and strength (Jeyaraj and Zadeh, 2020). Flexibility in operations and reduced cycle time are core components of the digitalization of SC (Stank et al., 2019). In addition, the availability of more data on a real-time basis helps in quick decision-making by reducing uncertainty in SC. The literature on SCI deals with the social and technical aspects.

Further, equally important information sharing and maintaining effective communication among SC partners are also addressed (Tseng and Liao, 2015). Various types of technologies used in the SCD process are shown in Table 2. Block chain, cloud computing, machine learning, big data analytics, drones, the internet of things, advanced robotics, 3D printing, cyber- physical systems, additive manufacturing, virtual and augmented reality and artificial intelligence are the prominent technologies used in the digitalization process.

Digitalization creates value opportunities and savings for companies (Ehret and Wirtz, 2017; Grubic and Jennions, 2018). The DSC also provides flexibility in business processes and economies of scale (Gerpott and May 2016). Integration of the SC can be done using digital technologies with the help of IoT theories and physical systems (Luz Martin-Pena et al., 2018). Mendhurwar and Mishra (2021) identified opportunities related to social and IoT technology integration and challenges associated with the convergence of technology, focusing on cybersecurity challenges.

SC and the risks involved can be managed effectively with digital technologies. Further, digital technologies facilitate redesigning complex data-driven systems to understand how various SC aspects are interrelated. As a result, organizations can plan for initiating advanced actions to enhance the performance, reliability and efficiency of SCs.

4.3. Conceptual framework for implementation and operational dimensions of SCD

The central aspect of SCD is the intensity of information sharing between the partners. Information and communication technology (ICT) tools provide a better platform for coordinating and participating in SC activities. Moreover, long-term partnership in SC is fruitful only when the strategic information flow and trust are established among the partners. In recent times, organizations' strategies for SCD have changed drastically with the emergence of the state of the art digital technologies.

Digital technologies help real-time information transmission and support knowledge management practices (Wilkesmann and Wilkesmann, 2018). Adopting digital technologies increases customer databases, enhances SC performance, and helps build an end-to-end SC that expedites the processes. After the systematic review of 102 research papers, five key research elements were identified, and a framework was developed.

The major elements identified were related to the implementation and operational dimensions of SCD. The five major elements are (i) requirement analysis (Seyedghorban et al., 2020), (ii) implementation process aspects (Zangiacomi et al., 2020), (iii) tools and techniques (Hennelly et al., 2019), (iv) smart solutions and business models (Luz-Martin Pena et al., 2018) and (v) decision and performance management (Ivanov, 2020). A framework exhibiting the implementation and operational dimensions of SCD is shown in Fig. 7.

The measures of SCD are information sharing and interdependence among SC members. Information sharing, decision-making at the inter-organizational level and planning among SC partners are considered the key elements. The other key dimensions identified for SCD are information sharing and operational coordination, collaboration and information sharing, and physical integration. It is seen that most of the research

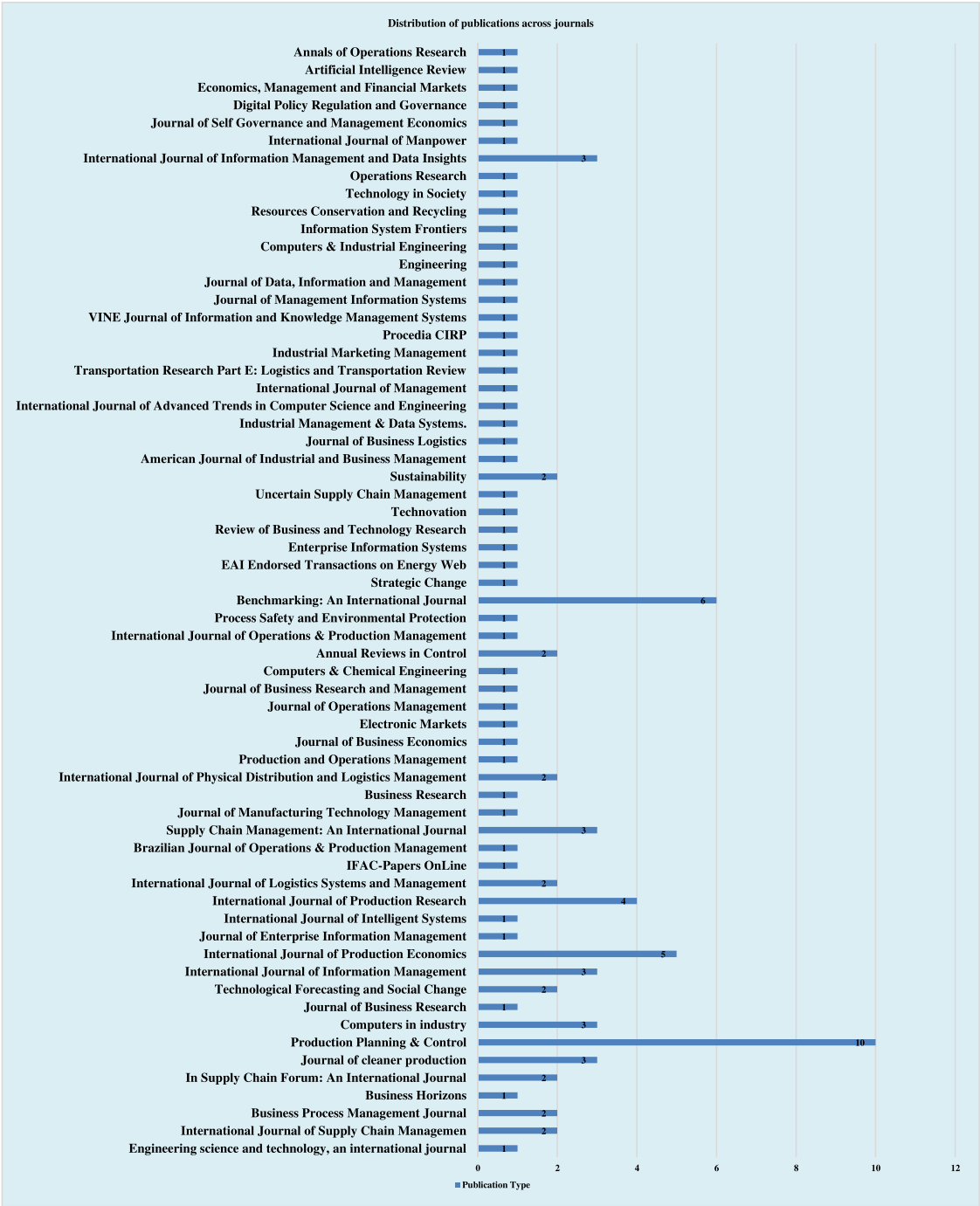

Fig. 4. Details of the distribution of articles across various journals.

papers in SCD have focused on the two dimensions of internal and external integration. Queiroz et al. (2019) developed a DSC framework consisting of seven basic capabilities and six enabler technologies. Digital technologies play a dynamic role in effective SC functioning and enhancing a firm's performance. There is a positive relationship between SCI and the firm's performance by comprising information, operational and relational integration. Based on the framework developed on the operation and implementation dimensions of SCD, each framework element is extracted and explained in the subsequent sub-sections.

4.3.1. Requirement analysis

The digitalization process needs to consider many functional areas, such as supplier relationship management, material requirement planning, enterprise resource planning, customer relationship management etc. In recent years, the digital transformation of SC using advanced ICT tools has gained greater significance. The research areas include human-machine interactions, conceptual articles on Industry 4.0, machine-equipment interactions, sustainability and technologies of Industry 4.0. Sustainable Industry 4.0 is developed with three main components: sustainable outcomes, industry 4.0 technologies, and process integration (Kamble et al., 2018).

The development of information technology (IT) has transformed the structure of SC by bringing organizations closer through collaboration and integration with SC partners (Chen, 2019). Real-time access to data enables the firms to plan production activities considering products in

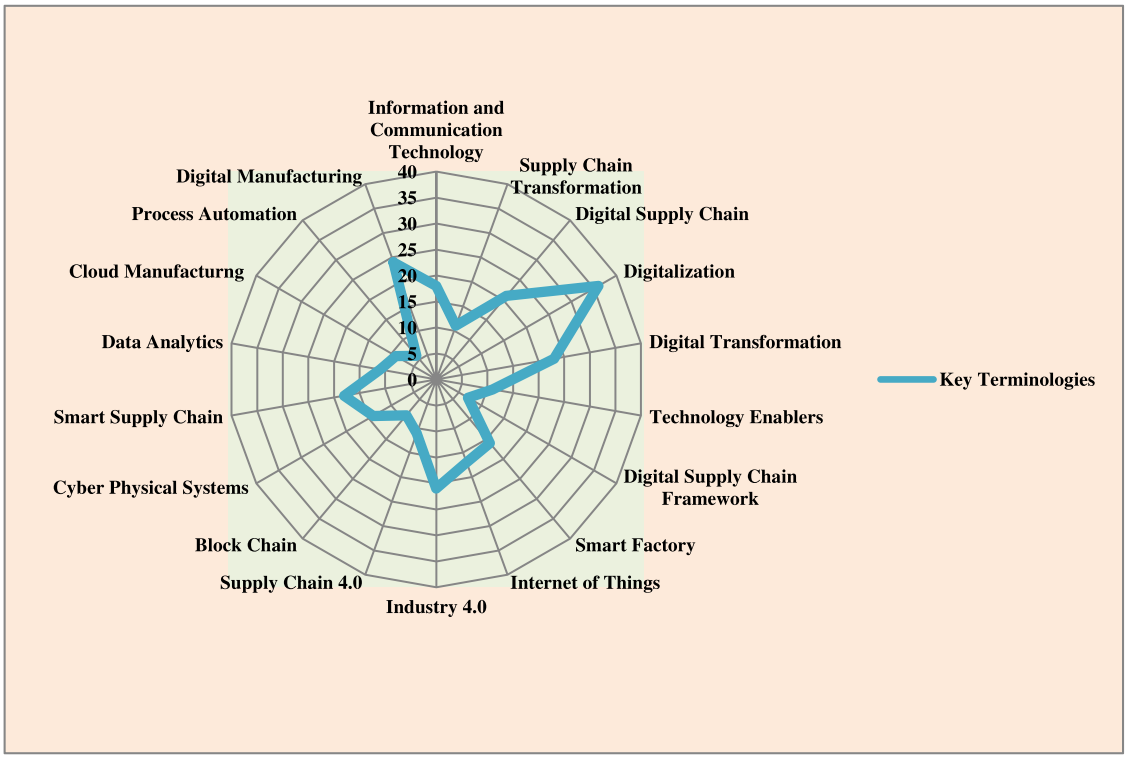

Fig. 5. Diagrammatic representation of key terminologies.

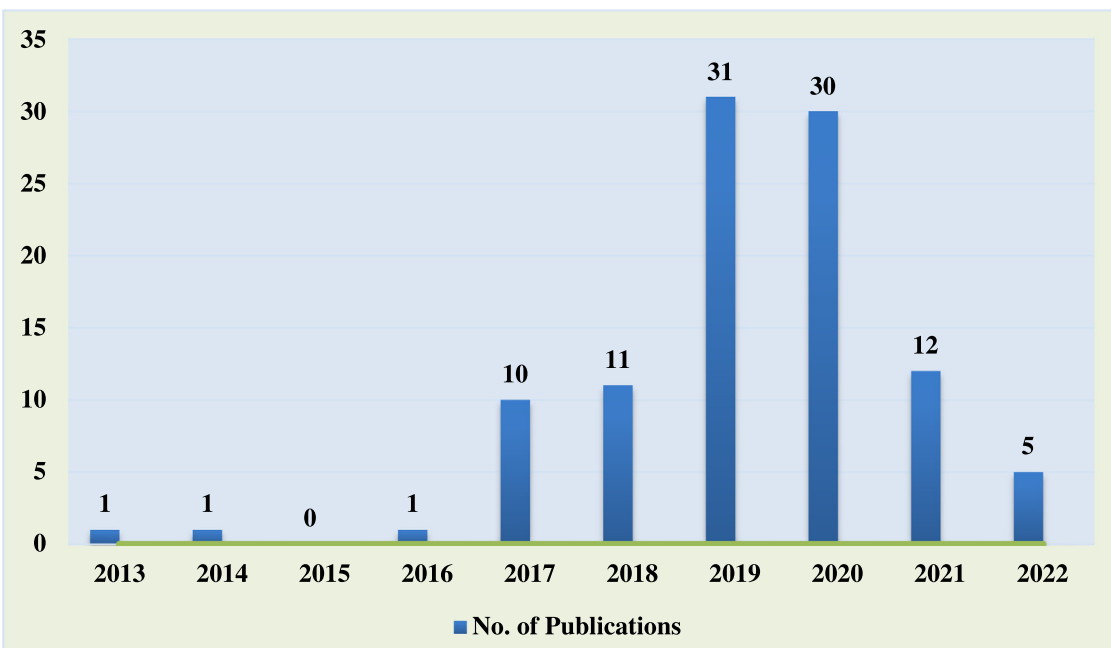

Fig. 6. Year-wise distribution of publications over time.

the pipeline and any anticipated problems. ICT tools and their application in planning, forecasting, and scheduling activities bring more transparency, visibility, accountability, and efficiency to SC. The impact of the revolution in SC due to emerging technologies is transforming businesses (Ben-Daya et al., 2019; Ivanov et al., 2019). The digitalization process in a company is a possible way to tackle uncertainties and customer demands effectively. I 4.0 and the concept of smart factories are gaining relevance recently (Liao et al., 2017).

Gupta et al. (2020) explored the orientation of firms in adopting Industry 4.0 and developed a theoretical framework based on institutional and dynamic capabilities view theories. They mentioned that only coercive pressure mediates a firm's intention to adopt Industry 4.0 and

Table 2
Various types of technologies used in SCD.

Sl. No.	Technology	Reference
1	Block chain	Hennelly et al., 2019; Attaran, 2020
2	Cloud computing	Zhong et al., 2017; Panitto et al., 2019
3	Machine learning	Min et al., 2019; Hennelly et al., 2019
4	Big data analytics	Armengaud et al., 2017; Brinch et al., 2021
5	Drones	Kittipanya-Ngam & Tan, 2020; Deja et al., 2020
6	Internet of things	Luthra et al.,2018; Boyes, 2018
7	Advanced robotics	Bechtsis et al., 2018; Ivanov et al., 2019
8	3D Printing	Seyedghorban et al. 2020; Kittipanya-Ngam & Tan, 2020
9	Cyber-physical systems	Panitto et al., 2019; Alcacer et al., 2019;
10	Additive manufacturing	Muthusami et al., 2018; Isaksson et al., 2018
11	Virtual and Augmented reality	Zhao et al., 2019; Zangiacomi et al. 2020
12	Artificial intelligence	Schniederjans et al., 2020; Grover, Kar, & Dwivedi, 2020

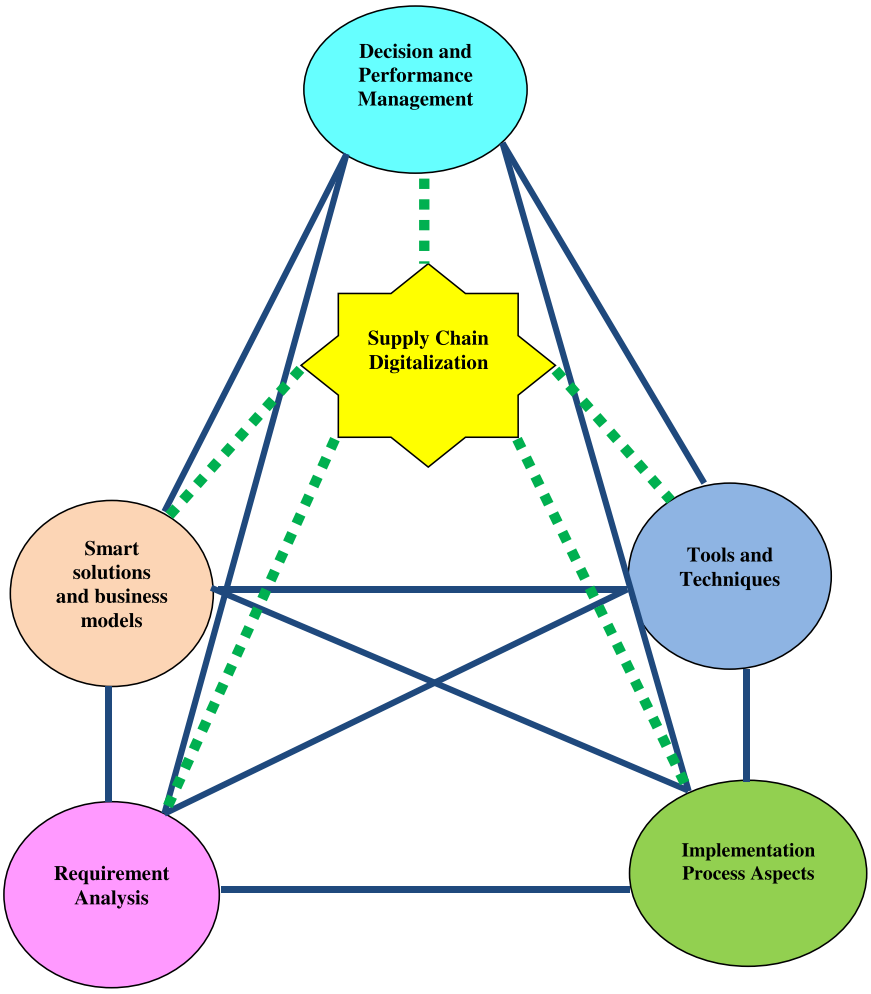

Fig. 7. A framework exhibiting the implementation and operational dimensions of SCD.

DSC. The application of novel technologies in the SCM processes of plan, source, make and deliver based on the Supply Chain Operations Reference (SCOR) model is seen in the literature. Ehie and Ferreira (2019) investigated how SCD influences SC capabilities and performance using the firm's resource-based view.

Hence, considering various aspects, the components of requirement analysis can be decomposed into four major elements, viz. (i) theoretical underpinning, (ii) dynamic capability assessment, (iii) digitalization trends and (iv) road map for digitalization. Literature reveals organizational information processing theory (Kamble and Gunasekaran, 2020), the resource-based view (Seyedghorban et al., 2020) and the dynamic capabilities view (Gupta et al., 2020) as the major theories relevant to the digitalization process. Dynamic capability assessment for the digitalization process can be initiated by assessing the existing capabilities (Brinch et al., 2021), compatibility with existing systems (2018) and supply chain stakeholder evaluation (Cenamor et al., 2017; Wamba and Queiroz, 2020).

To assess the trends in digitalization, aspects such as competitor assessment (Vendrell-Herrero et al., 2017), transformation functions/areas (Zangiacomi et al., 2020), and integration aspects (Kittipanya-Ngam and Tan, 2020) have to be considered. Further, a roadmap for digitalization is to be finalized considering the digitalization vision (Stank et al., 2019), finalization of procedure (Sarkis, Kouhizadeh, & Zhu, 2020) and application of technologies (Ehie and Ferreira, 2019). A diagrammatic representation of the key components of digital requirement analysis is shown in Fig. 8. The main

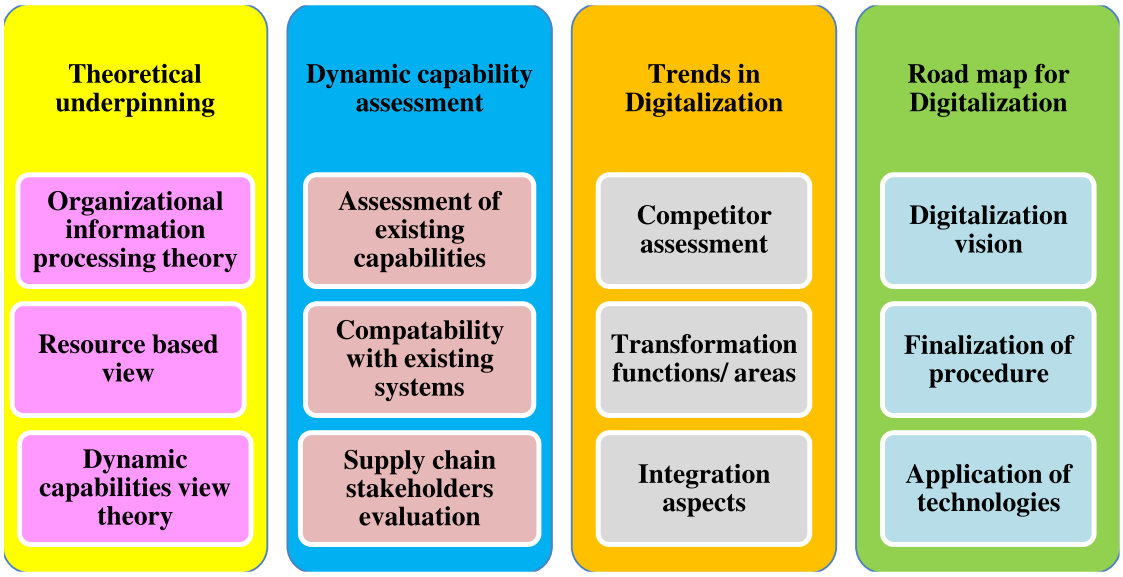

Fig. 8. Key components of digital requirement analysis.

components in analysing the digital requirement include identifying theoretical aspects of the digitalization process and assessing the dynamic capability. Accordingly, the existing capabilities must be assessed, scrutinizing compatibility with the existing systems and evaluating supply chain stakeholders' requirements. The digitalization trends are also to be analyzed considering functional areas that can be transformed, the competitor's capability and the feasibility of integration with the existing system. After this, the road map for digitalization can be finalized by considering the vision, procedure to be adopted and application of the required digital technologies.

Stank et al. (2019) introduced a digitally dominant paradigm based on theoretical underpinning to illustrate the infusion of digital concepts into the existing system of SC. They used middle-range theorizing to explain the SC process during the digitalization era. Brinch et al. (2021) proposed a holistic approach to firm-level capabilities that assists in creating value from big data. They identified significant capabilities like IT, performance, process, human strategy and organizational practices.

The management of operations and SC management challenges can be done effectively by applying control theory to provide better insights. Ivanov et al. (2018) examined the applicability of control theory in management and engineering problems and established the relationship between operations management and control systems theory. They also found that effective collaboration between the SC experts and control engineers would assist in introducing enhanced and better practically viable models, resulting in a competitive advantage in production and Industry 4.0 networks. Kunkel et al. (2022) explored SCD in the electronic industry and its implications for sustainable supply chain collaboration.

The application of ICT in small and medium enterprises can be leveraged based on concepts like the requirement of human resources specialized in IT, the relationship between buyer and supplier, knowledge-sharing activities, and acceptance of electronic markets (Scuotto et al., 2017). Tortorella et al. (2020) studied the mediating role of learning capabilities in an organization at different contextualization levels between digital technologies and operational performance. They found that a higher operational performance level can be achieved by mediating organizational performance and its impact on Industry 4.0. Digital transformation in logistics and SCs can be achieved through systematic and effective organizing of the activities. The beginning of SCD in the global scenario can be done through big data (Marmolejo-Saucedo and Hartmann, 2020).

4.3.2. *Implementation process aspects*

Samadi and Kassou (2016) investigated the IT productivity paradox in the performance of the SC through the SLR of 33 papers. They identified the indirect impact of IT on SC performance as their significant finding. Adopting the industrial IoT leads to industrial automation and control systems that provide enhanced connectivity to industrial systems. The relationship between the industrial IoT and concepts like Industry 4.0 and cyber-physical systems have been reported in the literature.

Smart solutions based on emerging technologies make complex systems vulnerable to security risks and the non-availability of suitable IT components. Hackel et al. (2019) developed a risk assessment model for the investment decision-making process in the IT security aspect using graph theory, matrix notation and value at risk. The process of integrating digitalization into operations and SCM and its feasibility and maturity are seen in the literature. Han et al. (2021) analysed the challenges of SCD in various business environments and evaluated the supply chain performance from the perspective of quality and cost.

Kumar, Singh, & Modgil (2020) explored the issues in the agri-food SC due to the influence of ICT and its impact on SC performance. They found that IT and SCM practices positively influence a firm's performance. Majeed et al. (2017) gave a conceptual framework for enhancing the inbound and outbound operations in the fashion apparel and footwear industries. They used a mixed-method approach to analyze how the transactions can be executed effectively in the system. Mendhurwar and Mishra (2021) identified the interactions among digital technologies, synergetic relationships and challenges on security issues.

The key components of the implementation process aspects of SCD include the identification of digital technology enablers, adoption challenges, information security risks, disruption, and strategies. The element of digital technology enablers includes organizational enablers (Bordeleau et al., 2020), technological enablers (Samadi and Kassou, 2016) and investment assessment (Hofmann and Rusch, 2017). Standardization (Kumar, Singh, & Modgil, 2020), adaptability (Kittipanya-Ngam and Tan, 2020) and interoperability (Majeed et al., 2017) constitute the adoption challenges elements. Information technology security and risks include cyber security risks

Fig. 9. Key components of implementation process aspects.

(Hackel et al., 2019), lack of technical skills and data security and privacy (Mendhurwar and Mishra, 2021). The disruption and strategies include extending disruption assessment finalization of strategy (Wilkesman and Wilkesmann, 2018) and strategy implementation (Panetto et al., 2019). Becue et al. (2021) explored the opportunities and challenges of applying artificial intelligence and cyber security threats in the context of I 4.0. A diagrammatic representation of the key components of implementation dimensions is given in Fig. 9.

The implementation process encompasses criteria like enablers of digital technology, challenges involved, risks and strategies in managing disruptions, etc. Considering digital technology enablers, the required organizational, technological, and extent of investment must be assessed. The challenges involved in adopting technology include the hurdles involved in standardization, its adaptability and interoperability. Information technology risks include cyber security, lack of technical skills and data security and privacy risks. The components involved in managing disruption include the assessment of the extent of disruption and the finalization of strategy and its implementation.

Panetto et al. (2019) highlighted the significance of the requirement for the effective coupling of systems and evolving systems. Bordeleau et al. (2020) found that the resources from enterprises and capabilities alone are insufficient to foresee the business value. They opined that the business performance in manufacturing medium enterprises could be leveraged through business intelligence and analytics.

Hofmann and Rusch (2017) presented the opportunities for logistics in the context of Industry 4.0 and its implications. They opined that Industry 4.0 has the potential to influence various industry functions, including designing goods, production, delivery and payment (Hofmann and Rusch, 2017).

Kittipanya-Ngam and Tan (2020) developed a digitalization framework by studying the opportunities, practices, and challenges faced by food manufacturers in Thailand. Wilkesmann and Wilkesmann (2018) proposed various theoretical frameworks to analyze the application of Industry 4.0. They found that applications of industry 4.0 vary based on dimensions of the degree of formalisation, location of control authority, knowledge and degree of professionalism.

4.3.3. Tools and techniques

Kache and Seuring (2017) analyzed the potential impact of big data analytics on digital technologies in the SC. Also, they presented various challenges and opportunities related to big data analytics in SC. Queiroz et al. (2020) proposed an integrated framework consisting of big data analytics, cyber-physical systems and IoT with 26 drivers related to smart production systems for improving business process management. Their study emphasized the role of the smart production syste, IoT, business process management, cyber-physical systems, and improvement in business process management. Kushwaha et al. (2021) studied the application of big data in various upcoming management disciplines.

The tools and techniques applicable to SCD include various methods used for production, like additive manufacturing (Frank et al., 2019), smart manufacturing (Alcacer and Cruz Machedo, 2019) and machine learning (Sarkis et al., 2020; Lee and Lim 2021). The major tools that can be considered include virtual and augmented reality (Buchi et al., 2020) cyber-physical systems (Queiroz et al., 2020) and advanced robotics (Tang and Veelenturf, 2019; Mir et al., 2020), Block chain (Pan et al., 2020), artificial intelligence (Tang and Veelenturf, 2019; Malik, Tripathi, Kar, & Gupta, 2021) and cloud services (Apiyo and Kiarie, 2018) come under the technologies element. The analytics part includes cognitive analytics (Hastig & Sodhi, 2020), big data analytics (Kache and Seuring, 2017) and business analytics (Hausberg, 2019). A diagrammatic representation of the key components of tools and techniques is given in Fig. 10.

The key components of tools and techniques include assessment of production techniques, tools, technologies and analytics. The prominent techniques related to production include additive manufacturing, smart manufacturing and machine learning. The major tools and technologies considered in the digitalization process include virtual and augmented reality, cyber-physical systems, advanced robotics, block chain, artificial intelligence and cloud services. The analytics part includes cognitive analytics, big data analytics and business analytics.

The business models, production systems, machines, environments, operators and products and services are becoming digital through interconnected systems with corresponding virtual representation. Alcacer and Cruz-Machado (2019) highlighted the industry 4.0 environment and the enabling systems and technologies and systems used in the manufacturing environment. Apiyo and Kiarie (2018) stressed the role of ICT tools in accessing real-time information and effective interactions between various organizations and the SC to gain a competitive advantage. Different decision support models for SC in production planning and scheduling, logistics planning and demand management are available. Major technologies used in the digital era include additive manufacturing, data analytics, sensors, blockchain, augmented reality, DT., etc.

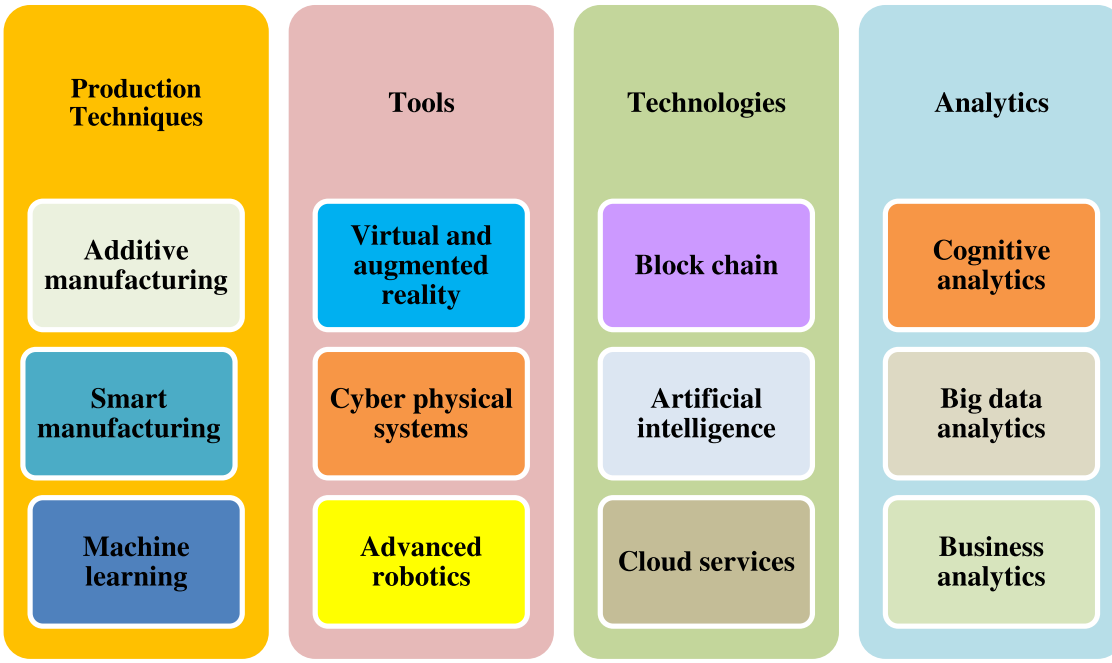

Fig. 10. Key components of tools and techniques.

The relevance of analytic culture in managing organizational change is seen in the literature. Business analytics and cognitive analytics play vital roles in the procurement process. Handfield et al. (2019) have proposed elaborated strategies for implementing advanced technologies for procurement. Various disciplines of digital transformation research identified nine major areas overlapping with the less explored areas. Innovation management, marketing and finance are the core business research area identified. Hausberg et al. (2019) have found that research in digital transformation in sustainability, human resource management and accounting is still lacking.

Frank et al. (2019) have explained the pattern of adoption of digital technologies in manufacturing firms based on a conceptual framework, considering the base and front-end technologies. They regarded smart manufacturing, smart SC, smart products and smart working in the front-end technologies, while IoT, big data and analytics and cloud services as base technologies.

A measurement model for practical work and a base for future research on blockchain applications for SC traceability is also seen in the literature. The traceability systems are to be employed by identifying various requirements and critical factors during the implementation process. The capability of companies, maturity of technology, collaborative SC practices, governance of traceability efforts and leadership were identified as the critical factors (Hastig & Sodhi, 2020). Wamba and Queiroz (2022) have developed a model to determine the block chain diffusion across supply chains. The diffusion was analysed based on the resource-based view, technology adoption model, institutional theory dynamic capability and innovations theory. They also explored the essential differences in the variables that determine blockchain innovation across countries.

Pan et al. (2020b) have done a quantitative analysis of 50 listed blockchain technology companies in China and found that the expansion of enterprise asset scale is a significant driving factor for implementing blockchain technologies. They also reveal that reduced sales expenses and enhanced asset turnover rates are influenced positively by blockchain technology. Sarkis, Kouhizadeh, & Zhu, 2020 highlighted the role of digitalization in achieving environmental SC sustainability by considering possible trade-offs of greening and digitalization. They also developed frameworks for traditional and emerging digitalization and IT.

Buchi et al. (2020) analyzed the causal relationship between the degree of openness of a firm to the technologies of industry 4.0 and its performance through a case study in manufacturing firms. They measured the degree of openness based on the number of digital technologies used and the depth of value chain stages involved. Cyber-physical systems can be developed by leveraging digital technologies to gain a competitive advantage. New technologies include additive manufacturing, artificial intelligence, advanced robotics, IoT, blockchain, autonomous vehicles, drones etc. (Tang and Veelenturf, 2019).

4.3.4. Smart solutions and business models

The "Supply Chain 4.0" terminology was proposed based on a conceptual framework by capturing Industry 4.0 essence. Luz Martin-Pena et al. (2018) have mentioned digitalization and servitization in achieving competitive advantage in manufacturing firms through innovative digital business models. The enhanced competition has made most manufacturers look for customized models through servitization.

Research on how digital platform owners gained a competitive advantage gets mentioned in the literature. Hein et al. (2019) proposed a framework integrating the strategies at the firm level and environmental factors in gaining market dominance. Organizations can generate valuable information by effectively managing knowledge based on generated data. Knowledge management can improve the research paradigm in SCD, optimizing the DSC's performance (Schniederjans et al., 2020).

The process of digitalization results in the creation of smart solutions and digital platforms and leads to industrial automation. Smart supply chain (Heinen and Hoberg, 2019), smart factory (Cenamor et al., 2017) and smart working (Zhong et al., 2017) are some of the smart solutions. The digital platforms include digital business models (Ardito et al., 2019), modular architecture , and state-of-the-art factories (Di Vaio and Varmala, 2020). Industry 4.0 (Zutshi and Grilo, 2019), DSC Model I 4.0 (Sundarakani et al., 2019) and SC 4.0 ((Luz Martin-Pena et al., 2018) form the industrial automation element. A diagrammatic representation of the key components of smart solutions and business models is given in Fig. 11.

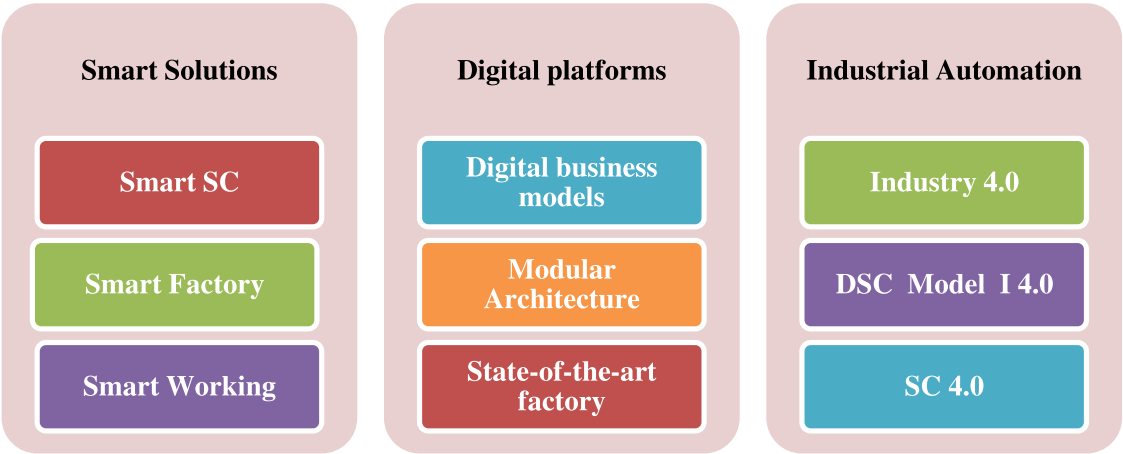

Fig. 11. Key components of smart solutions and business model.

The smart solutions include smart SC, smart factory and smart working. State-of-the-art factories, modular architecture and digital business models are the platforms considered in the digitalization process. Industrial automation includes Industry 4.0, DSC model I 4.0 and SC 4.0.

Cenamor et al. (2017) explored a platform approach to implement digital technologies for service offerings in manufacturing firms. They highlighted the impact of digital and information technologies on enhanced service offerings. They found that a platform approach based on modular architecture enables manufacturers to attain operational efficiency and customization. Integrating intelligent autonomous vehicles is one of the emerging fields in sustainable SC networks. Accordingly, a framework that captures the key software architecture elements in developing customized simulation tools is found in the literature. Bechtsis et al. (2018) highlighted the dynamic reconfiguration of the SC network based on the flexibility of decentralized software management architecture.

Literature also records the implications of digital platforms in redesigning and rethinking various processes in business, considering the inter-organizational relationship among SC partners. Adopting digital platforms results in benefits like reducing schedules and documentation in processes (Di Vaio and Varriale, 2020). Zekhnini et al. (2020) developed a road map framework by identifying the influence of innovative technologies on SC networks. They outlined the components for SC transformation and identified the barriers, motivations and impact of technology in various SC processes. Kumar, Vrat, & Shankar, 2021 have analyzed barriers and prioritized various strategies to overcome the barriers in Industry 4.0.

The flexibility in manufacturing can be enhanced by adopting Industry 4.0 practices, resulting in better productivity, quality, and mass customization. Zhong et al. (2017) comprehensively analyzed intelligent manufacturing, cloud manufacturing, and IoT-enabled manufacturing. Ardito et al. (2019) elaborated on the role of digital technologies in acquiring information, elaboration, and storage for SC integration. They also examined the potential of IoT for production, business and operating models. Boehmer et al. (2020) presented theoretical propositions like the impact of the implementation of digital technology on the exchange of information and sharing of financial and risk incentives.

Data analytics are used in digital manufacturing to analyze the risk profiles of the supplier's performance in uncertainty. Cavalcante et al. (2019) proposed an approach for resilient supplier selection using a hybrid machine learning and simulation approach in creating digital twins and their applications to data-driven decision support systems. They theorized and elaborated on the risk profile of the supplier's performance and resilient SC performance. Various managerial and technical issues affect the smart manufacturing system based on 3D printing. Chen and Liu (2017) have found that technical problems should be addressed to safeguard the viability of 3D printing applications. They believed that for optimizing and enhancing the 3D printing applications, managerial challenges should be resolved.

Additive manufacturing allows for switching from traditional manufacturing by considering cost aspects, production process and operations management. Heinen and Hoberg (2019) followed a universal perspective approach in developing a design to analyze the impact of additive manufacturing and also evaluated it in a case company.

Digital platform business models have led to optimizing efficiencies and enhanced information sharing capabilities. Cloud computing enhances competitiveness through cost reduction, flexibility, and enhanced elasticity through optimum utilization of resources. Sundarakani et al. (2019) developed a SC cloud model for transforming the partners towards Industry 4.0.

To cope with technological advancements, businesses have been forced to redesign their strategies. Zutshi and Grilo (2019) have highlighted the importance of digital platforms and their effect on industrial engineering. They elaborated on their study's essential aspects, characteristics and value proposition of digital platforms. Joshi and Gupta (2019) have developed an advanced production system using IoT that satisfies various components, products and materials demands. The products' designs are evaluated for remanufacturing and disassembly in their approach, considering multiple criteria like total profit, number of disposed of items, and the quality level.

4.3.5. Decision making and performance management

Attaran (2020) identified the implementation and managerial challenges for adopting and implementing digital technologies. Digital models of products and customers are used in digital product fitting to meet customer requirements.

Decision making, performance management, risk management and knowledge management form the major components of the decision making and performance management category. Under the decision making component, data-driven visualization (Ishtiaque et al., 2020), business intelligence (Chen et al., 2019) and data sources outlets connectivity (Dunke and Nickel, 2020) are included. Performance management comprises openness (Ukko et al., 2020), integration of SC solutions (Isaksson et al., 2018) and enhanced collaboration (Nasiri et al., 2020). Risk management (Attaran, 2020), risk assessment analytics (Ivanov et al., 2019) and risk mitigation (Xue et al., 2013) come under risk management categorization. Knowledge management includes knowledge creation (Ardito et al., 2019), knowledge storage and transfer (Zangiacomi et al., 2020) and knowledge application (Schmiederjans et al., 2020).

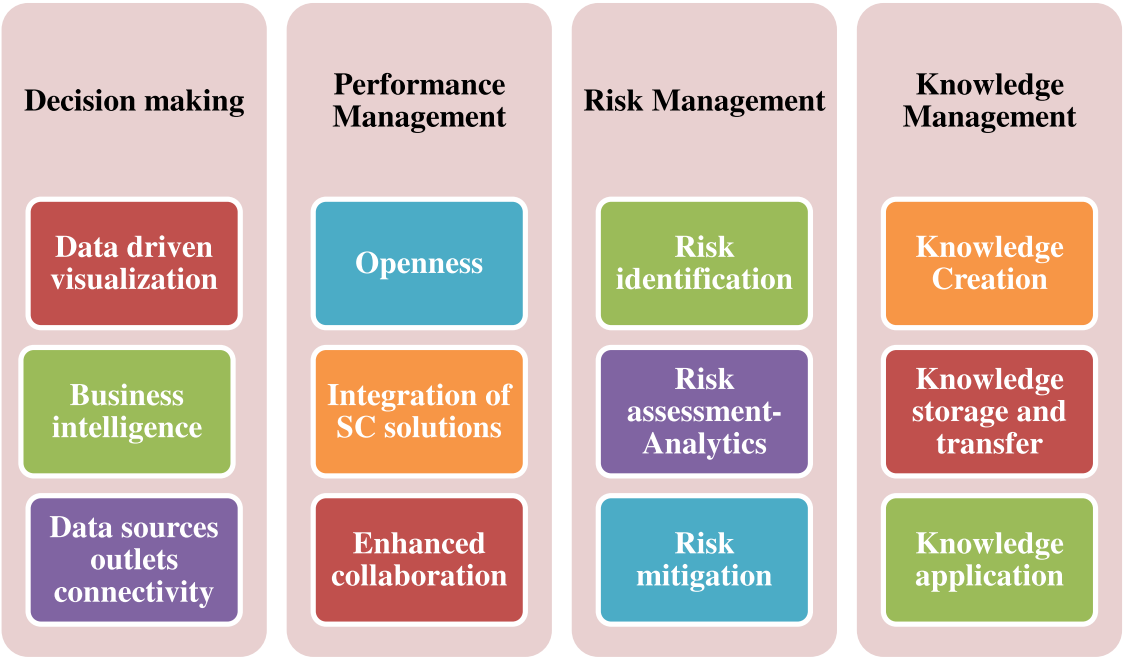

Fig. 12. Key components of decision making and performance management.

IoT leads to the development of new system architectures. Its application leads to improved connectivity facilitating real-time availability of information and effective integration of isolated functions. The reliability of modeling at the right level is a significant factor in success (Isaksson et al., 2018). Ivanov et al. (2019) have studied the impact of digitalization on the ripple effect and risk control analytics in the context of Industry 4.0 transformation in SCs. They investigated the relationship between business, engineering, information and analytics on SC risks and digitalization. Modular systems theory has been used to explain risk mitigation while adopting the digitalization of SC systems. The assignment of decision rights to the IT unit enhances the risk-mitigating effect on system modularity (Xue et al., 2013). A diagrammatic representation of the key components of decision-making and performance management is given in Fig. 12.

The key components of decision making and performance management include data-driven visualization, business intelligence, and connectivity to data sources outlets. Enhanced collaboration, integration of SC solutions, and openness are important components in performance management. Risk management includes risk identification, risk assessment analytics, and the risk mitigation process finalization. Knowledge creation, storage, transfer and application are the key components of knowledge management.

Digital transformation in the SC results in enhanced relationship performance through effective coupling with smart technologies. Thus, the approach of mediating SC's competitive advantage and performance can be made through smart technologies (Nasiri et al., 2020). Ukko et al. (2020) found the factors affecting the relational mechanisms of measuring performance in digital service SC. Their research has emphasized trust among SC partners and enhanced information sharing as the main relational performance measurement mechanisms.

A novel research model explaining the relationship between information sharing, SC integration, business performance and operational performance is cited in the literature. Chen et al. (2019) have found that information sharing is crucial to improve SC integration and operational performance. They stress that the influence of information systems on organizational performance would provide better management insights for developing suitable strategies.

Alsufyani and Gill, 2022 studied the impact of digitalization on organization performance outcomes and analyzed the results using adaptive enterprise architecture. The literature shows the applications of optimal control in production scheduling, Industry 4.0 and SC by focusing on the deterministic maximum principle. Dolgui et al. (2019) highlighted the models for control engineering with production management and their linkage with industrial engineering and management. Dunke and Nickel (2020) explored collaborative IT applications in enhancing intra-organizational logistics. They reported that drastic improvement was gained after adopting collaborative IT solutions.

The most effective way to enhance the performance of SC operations is through adopting ICT, which improves integration capabilities (Ishtiaque et al., 2020). The performance measures and metrics can enhance the firm's performance based on effective communication among various SC partners in the network. The metrics need to be developed considering two major categories: big data analytics capability measures and assessing the performance of big data-driven SC processes.

Zangiacomi et al. (2020) analyzed the managerial perspective for digital transformation leading to Industry 4.0 through multiple case studies in the manufacturing value chain. They identified three significant dimensions: investments in digital technologies, knowledge sharing and the ability to perceive digitalization. They also proposed the key challenges and appropriate practices in digital implementation.

It is found that research on the implementation, benefits and design of SC is scarce (Hofmann & Rusch, 2017). Ardito et al. (2019) have addressed the innovative efforts to develop digital technologies as an interface for marketing and sustainable SCM from an information processing perspective. It provides insights into the application of digital technologies in enabling SCM marketing integration. The transformation process towards I 4.0 and Industry 5.0 includes applying advanced digital technologies and integrating SC performance. In this context, enterprise architecture is the ideal platform that can be used to enhance the flow of information and SCI.

IOIS is a value-added shared information system between the participants in the business process that facilitates generating, transforming, communicating, and storing information across the firm's boundaries. Deepu and Ravi (2021) explored the application of Inter-Organizational

Information Systems for SCD. Gupta et al. (2020) have addressed the orientation of firms in adopting I 4.0, and DSC assists in decision making. The emerging research streams and gaps in the literature on SCD are presented in the next section.

5. Emerging research streams and gaps in the literature

Based on the detailed literature review, we propose future research directions to generate innovative knowledge and theories in SCD. The study also recommends further research into real-case industrial applications based on the proposed framework. Accordingly, the following areas for research are presented. SCD has become vital for enterprises to integrate their SC effectively by devising suitable strategies. A digitally integrated SC will help timely exchange information across partners, resulting in cost reductions and better manufacturing efficiency. The digital transformation process influences various disciples in research and practices in the organization leading to specialized and independent streams in literature. Hausberg et al. (2019) have presented a framework considering nine core areas that assist the implications of future areas of research based on digital transformation.

The sustainability functions of Industry 4.0 have been mentioned by reviewing the trends in technology and basic design principles of Industry 4.0. It is found that complex relationships exist among different sustainability functions of Industry 4.0 (Ghobakhloo, 2020). The evolution of a connected ecosystem based on Industry 4.0 and framing strategies to become transparent and competitive is mentioned in the literature. Seyedghorban et al. (2020) have applied quantitative and qualitative techniques to classify the literature.

Various aspects suggested for future research include humanizing manufacturing through digital manufacturing strategy, data science-enabled SCM, SC agility, omnichannel, IoT, and beyond. Panetto et al. (2019) propose future research areas as modeling in manufacturing, management and control based on the innovative production and resilient SC networks in Industry 4.0. Considering these aspects, the emerging areas impacted due to digitalization and the gaps in the literature are mentioned in the following sub-sections.

5.1. Resilience through digitalization

Resilience refers to the ability of SCs to prevent and absorb changes and recover to their initial performance. Stakeholders should foresee the disruptions and devise suitable strategies to respond to the changes as quickly as possible. SC's resilience is a significant component of effective SC performance, achieved through efficient communication through digitalization. Digitalization allows the creation of a more responsive, transparent and agile SC network for effective planning and integration. The historical data and smart digital solutions can predict disruptions and changes to adjust the SC. Considering this, the following research gaps are identified for the effective implementation of SCD.

Gap 1. *The role of digitalization and the impact of various digital technologies in attaining SC resilience needs further investigation. SC disruptions caused due to the pandemic, their impact on resilience factor and the relative significance of digital technologies in the processes need to be explored further. The application of digital tools and their operationalization in measuring the extent of resilience in the SC needs deeper insights.*

5.2. Sustainability and sustainable manufacturing

The process of SCD can leverage SC sustainability (Sarkis, Kouhizadeh, & Zhu, 2020). I 4.0 and sustainability are interlinked. They facilitate efficiency enhancement of processes, optimum utilization of resources, better utilities for customers, and reduce pollution & waste. The managers must integrate digital technologies to achieve the sustainability purview's social, economic and environmental objectives. The benefits of digitalization and greening have been highlighted in the literature. Sustainability can be achieved by using eco-friendly materials for production, which are dependent on organizational structure. Ferasso et al. (2020) highlighted using advanced digital technologies to enhance productivity and reduce wastage. The sustainability approach facilitates developing and adopting methods like remanufacturing, recycling, recovering, reusing, rethinking, and reducing (Bag et al., 2021). Applications of various digital technologies like IoT sensing networks, AI-based decision-making algorithms and process monitoring on a real-time basis in sustainable I 4.0 were found (Nica and Stehel., 2021). Konstantakopoulos et al. (2021) have developed an algorithmic approach for collaborative and sustainable logistics. Kar et al. (2022) explored the impact of AI on sustainability. Adopting a sustainable manufacturing process allows cost reduction, optimum utilization of resources, enhanced brand reputation and regulatory compliance, increased market access, and competitive advantage. The following gaps in the literature on sustainability and sustainable manufacturing are proposed, which need further exploration:

Gap 2. *The impact of digital technologies and the role of each of the technologies in attaining sustainability needs to be explored. Advancements in SC due to technological impact, adoption of smart contracts, blockchain-based big data and the internet of things, artificial intelligence needs to be assessed further. The feasibility of creating new models for sustainability, considering the advancements in digital technologies, needs further investigation.*

5.3. Intertwined and viable SCs

An intertwined supply network (ISN) consists of interconnected SCs with dynamically changing structures. Tang and Veelenturf (2019) have highlighted the significance of I 4.0 and cyber-physical systems and the increasing intertwining of SCs. Furthermore, various SC interconnections in the sharing and circular economies are highlighted in the literature(Choi, Feng, & Li, 2020). Viability refers to the ability of the system to meet the requirements in a changing environment. SC viability has gained significance in operations management due to the outburst of the COVID-19 pandemic (Ivanov, 2020). In recent times, the disruptions in SC have stimulated practitioners and researchers to investigate further the aspects of resilience and viability, which ensures normal operations during disruptions (Azadegan & Dooley, 2021). Considering the significance of addressing disruptions and uncertainties, the following research gaps are identified in viability and intertwined SC that needs further investigation:

Gap 3. *The characteristics of intertwined SCs, strategies, digital technologies maturity, risk and SC performance management are the areas to be explored further. Development of these areas concentrating on theoretical applications in the background of digital technologies impact needs further research. A deeper understanding of the digital transformation resulting in the creation of intertwined SCs, implementation issues, challenges and strategies are also required. Another gap identified is the finalization of the methodology for selecting technology and solution providers for digital transformation.*

5.4. Servitization

Servitization refers to the shift from product-based business models to service-oriented ones by providing value-added services ranging from essential add-on services like spare parts and repair services. The role of cloud-based solutions and their support in remote monitoring and support for servitization at various business levels in the manufacturing sector were found in the literature (Hao et al., 2020). While adopting servitization, firms must effectively manage and adapt the SC network, which is challenging. In the case of manufacturing firms, servitization has been considered an outcome of manufacturing firms' internal factors (Kim and Toya, 2019), demand and revenue-based motivations, and manufacturing and service capabilities. In addition, the role of SC partner's relationship with stakeholders in providing integrated solutions and maintaining relationships with customers (Cenamor et al.,

2017) is highlighted in the literature. The impact of digital disruption on business-to-business interdependencies and the intersection of servitization, digital business models and SCM are addressed in the literature. During digital servitization, the upstream firms should use the resources to manage their strategic position in the SC network (Vendrell-Herrero et al., 2017). The following gaps in the literature on servitization are identified considering the significance of digitalization in SC:

Gap 4. *Better conceptualization of servitization, theoretical development, and its underpinning and analysis in the background of advanced digital technologies in real-world case scenarios need further investigation.*

5.5. Circular and shared economy

The circular economy has become one of the buzzwords in recent times. This topic is also relevant among industry practitioners in planning SC strategies. It considers various factors that can monitor resources and reduce waste and resource consumption by reusing existing materials. Pan et al. (2015) have studied the circular economy implementation strategies from the SC perspective. An economic system wherein the assets or services are shared among private individuals for free or at a cost. The disruptions in SC and the dynamic nature of SC demand in-depth analysis and deeper insights, leading to the following gaps in the literature:

Gap 5. *The concept of circular SCs and their impact on various functions need further exploration. Country-specific studies are also required to bring in a better understanding of the practices followed in various economies. SC disruptions and their impact in the background of the circular and shared economy may also be explored further. Conceptual and quantitative SC frameworks considering the shared economy concept are also required.*

5.6. Digital encapsulation

Digital encapsulation facilitates the easy flow of materials and goods across SC partners. ICT tools help share real-time information across the SC network for on-time decision-making. Perez-Lopez et al. (2019) have quantified the relationships among variables for adopting ICT in SCs. The drivers and enablers affecting the process of SCI have been investigated, with little consensus on the integration of SC through digitalization (Hausberg et al., 2019). Hence, further exploration is needed in digital encapsulation based on the following research gap identified:

Gap 6. *The role of emerging digital technology solutions in SC decision-making needs further investigation. The impact and application of digital technologies on various SC functions and data-driven decision-making need to be explored further. Value chain generation by restructuring SC based on cost reduction, minimizing bullwhip effects, data analytics, and improved logistics operations capacity building are other promising areas for future research.*

5.7. Inter-Organizational information systems

Heterogeneous enterprise information systems can be addressed through effective integration using Inter-Organizational Information Systems enabled synchronization (Xu, Zhang, Li, & Xu, 2015). An efficient SC network can be established by employing modern information-sharing systems and tools. Effective management of SC by using ICT tools aid in utilizing the firm's resources and capacity effectively. The innovations in ICT and its performance have enabled effective and efficient information systems to manage SCs effectively. Enterprise information systems are the main components that play a vital role in the I 4.0 era in realizing smart manufacturing systems (SMS). The limitations of traditional information systems are a lack of seamless integration, business intelligence, value-driven processes and dynamic optimization, lack of complete information, etc., as these satisfy few business needs. To effectively utilize the potential of inter-organizational information systems in executing SC functions, the following research gaps are to be addressed.

Gap 7. *The role of digitalization in attaining flexibility, agility and traceability in SC during disruptions is of great significance. The feasibility of developing various innovative strategies and business models to manage disruptions and risks considering advanced digital technologies needs further investigation. Strategies to address long-term impacts on SC disruptions based on digital technologies due to the pandemic may also be considered in the future research scope.*

5.8. Smart factory

Smart factory functions are enabled by considering cyber-physical systems, internet of services and the IoT by closely linking the machines, people, logistic systems and products. The smart manufacturing concept considers IoT, big data, cyber-physical systems, system integration, data analytics, sustainable manufacturing and additive manufacturing. Boehmer et al. (2020) explored the role of relational mechanisms of digital technologies in SC and have addressed the promising potential of IoT technology as an enabler for manufacturers in the production process. The wholly interconnected systems allow for identifying the real-time demand requirements, spare parts availability, location, and conditions that enhance cost efficiency. Sarkar et al. (2021) and Sarkar et al. (2022) analyzed the barriers and challenges in port logistics in emerging economies in the Industry 4.0 era. A fully integrated smart factory brings many more benefits, ultimately improving its profitability and controllability. The following research gaps are identified from the literature and presented.

Gap 8. *The impact of additive manufacturing and the potential benefits generated based on adopting digital technologies needs further investigation. The provision of sharing of capacity and digital technologies in the manufacturing process needs to be studied further. The role of adopting additive manufacturing and its relationship with SC resilience and SC sustainability may be investigated further.*

5.9. General aspects of digitalization processes

The benefits reaped from the digitalization process are an encouraging sign for the firms willing to proceed with SCD. However, the literature reveals that enterprises, especially in the manufacturing sector, lag during the pilot phase of digitalization. Thus, an appropriate roadmap for SCD has to be planned for effective digitalization and a proper investment return. The scope for further study into how knowledge management and digitization influence SC digital performance was also highlighted (Schniederjans et al., 2020). The current era leads to the digital and autonomous world through digitalization which forms the core area of SC. Research on various challenges and factors, interrelationships among the elements, framework and model for SCD are duly found in the literature. Thus, it becomes evident that there is still scope for further studies on SCD. Hence, the following gaps in the literature are also identified for further exploration on a broader basis.

Gap 9. *The application of DSC varies from industry to industry by adopting suitable methodologies using innovative digital technologies. Hence, the key trends of future DSC demand an industry-specific roadmap to enhance the digital transformation of SC-related activities. Furthermore, the framework developed can be applied to create industry-specific sub-frameworks to improve the knowledge base for academicians and practitioners.* Following are other promising areas and gaps identified from the literature on SCD:

- Studies on prioritizing customer and design requirements are necessary as organizations compete to transform by adopting DSC.
- A framework for SCD classifying the driving factors to fill the gap in existing theory through digitalization.
- To find out the theoretical gaps, conceptualization, and development of a model for IOIS adoption in SC.

- A model for quantitatively assessing the extent of digitalization through developing a digitalization index.
- Research into real-case industrial applications based on the conceptual framework developed and proposed emerging areas.

The analysis and findings of this study would be helpful to practicing managers to understand the current state of practice in SCD. It also assists in understanding the impact within industry and academia. The systematic review of the papers provides academicians and researchers with future research directions. The following section deals with the conclusion elaborating on this research's findings, implications and limitations.

6. Discussion

This section deals with discussions on the synthesis of literature, practical applications and future research directions.

6.1. Synthesis of literature

This study has adopted the SLR methodology to review the SCD publications and their analysis. The study is conducted covering a period of 10 years, spanned from 2013 to 2022. Based on the recent advancement in digital technologies, we feel the period considered is conclusive to cover the related literature on SCD. This research analyzes the evolution of SCD, and the constructs and significant areas that constitute SCD are elaborated. Further, the emerging trends and topics in SCD and the critical research gaps related to SCD are also identified. After the review of the literature, we, the authors, define SCD as follows.

"Supply chain digitalization can be defined as the process of integration and execution of supply chain functions using digital technologies considering the processes involved leading to the creation of smart solutions and business models for effective decision making and achieving better performance."

The findings are based on publications covering many highly-ranked academic journals, which can be considered comprehensive. An integrative framework for SCD aims at integrating the implementation of technology, advanced developments and SCM is developed. This research guides the professionals and managers in SC by providing insights for effective SCD by addressing the gaps. This review assumes significance as it has identified the key gaps in the literature by contributing to the theory of digitalization of SC. This study tries to bring out a theoretical background of digitalization in SC for its practical implementation and integration. The research further addresses what organizations must consider for digitalization to facilitate overall performance improvement of the SC. A detailed study on the research gaps identified in SCD is noteworthy to provide insights to practitioners and academicians to investigate the area further and contribute to the literature and practical applications.

This study proposed a robust framework for SCD as it has covered major dimensions to be considered in the digitalization of the SC. The framework also adds value to the practical application of the digitalization process. We have also identified the major research streams that emerged based on digitalization and its impact on SC activities.

This research aims to bring better insights into the process of SCD by systematically analyzing the articles published in the area of SCD. The significant contributions of this study are as follows:

- Firstly, this research has analyzed the evolution and extent of work done in the implementation and operational dimensions of SCD. The major components and dimensions involved in the implementation and operational dimensions of the SCD process were analyzed and presented. A conceptual framework is developed considering various aspects involved in the process of SCD.
- Secondly, the implementation and operational dimensions components were classified and grouped under major and sub-major areas in SCD. The bifurcated structure of the dimensions identified provides better insights into the process of SCD.
- Thirdly, the emerging trends and future research areas in SCD were also identified and mentioned.
- This review has also identified the key research gaps and future directions. Hence, the outcomes of this research have theoretical and practical implications.

Considering the theoretical aspects, this paper has identified and categorized the major dimensions of SCD. Further, the developed theoretical framework presented the multiple elements of digitalization processes and their outcomes. The essential research streams of SC were identified and mentioned, including gaps in the literature and future research directions. This will assist the research scholars in enhancing the knowledge base in the area of SCD. The research gaps identified also can contribute to the practical applications in the digitalization process and will further advance the knowledge of SCD and its practical applications.

6.2. Practical implications and future research directions

The key practical applications include real case applications in the area of digitalization based on the gaps identified; country-specific analysis of Industry 4.0 developments and its comparability with other countries; the influence of digitalization and its impact on new research streams like sustainability, viability, servitization, circular economy, resilience and intertwined SC. Practicing managers can address the challenges and risks involved in SCD to attain the real benefits of the digitalization process.

Further studies are required in various DSC areas based on the gaps identified in the digitalization process to provide dynamic, reliable and flexible SC. The feasibility of integrating with the existing SC can be done in real-time case applications to develop novel theories. SCD is a transformative process that considers all aspects of the digital era to explore its full potential. Accordingly, this research has highlighted a few critical areas that need further investigation. Even though the major aspects of SCD have been widely discussed from various perspectives, SCD is abundant with multiple research opportunities due to the advancement and developments in digital technologies. Further, combining SCD with other related research fields would enable research with a broader scope.

This research has certain limitations as well. The articles considered for the review were selected based on the keywords, abstract and title. Shortlisting of papers was done based on keywords. Thus, the non-consideration of publications in digitalization without the keyword is one of the limitations of this research. The scope of this study process is directly related to digitalization. This research has not considered the impact of digitalization on the development of emerging areas identified, which could be done in future studies. Researchers and SC practitioners can research the gaps identified in this study to develop new insights into the SCD process.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Appendix A
Literature review articles in the area of SCD.

Sl No	Author	Aim/ Objectives	Implications and Contributions
1	Samadi and Kassou, 2016	Relationship between IT and SC Performance	Literature review from 2000 to 2014 - 33 papers- Relationship between IT and SC performance needs more investigation.
2	Iddris,2018	DSC: Survey of Literature	Literature review from 2000 to 2017- 60 papers- Need to examine DSC development was highlighted.
3	Hausberg et al., 2019	Research Streams on digital transformation	Deepen the understanding of digital transformation in research.
4	Buyukozkan and Gocer, 2018	DSC: Literature Review and a proposed framework for future research	Literature review - 46 papers-Review of DSC, its importance and applications.
5	Hein et al., 2019	Digital Platform Ecosystems	Technology management, economics, and information systems have different perspectives on digital platform ecosystems.
6	Hackel et al., 2019	Assessing IT availability risks in smart factory networks	Structured approach and model for investment decisions on IT security measures.
7	Ehie and Ferreira,2019	Conceptual development of SCD framework.	Investigated the mechanisms by which SCD influences SC capabilities which in effect impacts operational performance
8	Apiyo, 2018	Role of ICT tools in SC Performance	Emphasized the embracement of ICT advancement and its operations in SC to stay ahead of the competition.
9	Attaran, 2020	Digital Technology Enablers and their implications for SCM	Examined the trends and challenges in DSC-Relevance of 5 G in industrial communications.
10	Garay-Rontero et al., 2020	DSC Model in I 4.0	They defined the main components of I 4.0 and their impact on DSC.
11	Schniederjans et al.,2020	SCD Trends- An Integration of Knowledge Management	Leverage knowledge management to understand the SCD process- Provides research inquiries to explore the ignored areas of SCD.
12	Kumar, Singh, & Modgil, 2020	Exploring the relationship between ICT, SCM practices in agri-food SC	Investigated the role of ICT in the agri-food SC; SCM practices significantly impact a firm's performance.
13	Zekhnini et al., 2020	SCM 4.0: a literature review and research framework	They identified the relationship between digital technologies and SCM.
14	Vendrell-Herrero et al., 2017	Servitization, Digitization, and SC Interdependency	They highlighted the role of ICT-enabled SCM practices in achieving competitive advantage.
15	Boyes, 2018	Industrial Internet of Things - An analysis framework	Industrial automation and control systems were isolated from conventional digital networks.
16	Naway and Rahmat, 2019	Technology and its relationship between SC capability and SC operational performance	Explored the link between SC capability and SC performance.
17	Wiedenmann et al., 2019	Literature review on the impact of digital technologies on operational causes of the bullwhip effect	Identified appropriate technologies for reducing bullwhip effects.
18	Sundaram et al., 2020	Digital Transformation of Business Models- A systematic review of the impact on revenue and SC	Impact on distribution models with integration of digital transformations- Need for incorporating digital shifts in business models.
19	Nasiri et al., 2020	Managing the DSC- The Role of Smart Technologies	Smart technologies mediate between digital transformation and relationship performance.
20	Zhao et al., 2020	Smarter SC- literature review and practices	How to address business, policy and technical challenges – Proposed successful practices and existing solutions for smarter SC.
21	Ukko et al., 2020	Relational mechanisms to performance measurement in DSC	Determined the relational mechanism of performance measurement in digital service SCs.

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Sl No	Author	Aim/ Objectives	Implications and Contributions
22	Marmolejo-Saucedo and Hartmann, 2020	Trends in the digitization of SC - a literature review	SC operations in a digital context- the evolution of SC in the digital context of operational functions.
23	Hennelly et al., 2019	Rethinking SC in the age of digitalization	Highlighted the importance of digitalization, its feasibility and its integration into the field of SCM.
24	Zhong et al., 2017	Intelligent manufacturing in the context of I 4.0- A review	New ideas for I 4.0 revolution - Worldwide movement in intelligent manufacturing, including effective plans.
25	Sorooshian and Tech, 2020	Information technology for SCM Literature Review	Listed the impacts of IT on SCM – Revealed a better understanding of IT adoption and their role in SCM.
26	Ivanov et al., 2019	The impact of digital technology and I 4.0 on the ripple effect and SC risk analytics	Digitalization to enhancing ripple effect control; Digital technology-based extensions that trigger developments towards SC risk analytics.
27	Stank et al., 2019	Toward a Digitally Dominant Paradigm for 21st Century SC Scholarship	Digitally Dominant Paradigm can impact SCM concepts - Application of Digitally Dominant Paradigm in Logistics/ SCM.
28	Handfield et al., 2019	Emerging procurement technology - Data analytics and cognitive analytics	Explored the current and future state of procurement analytics to enable the integration of analytical solutions.
29	Frederico et al., 2019	SC 4.0-Concepts, maturity and research agenda	Conceptualization of I 4.0 in the context of SC – Developed a conceptual framework for SC 4.0.
30	Muthusami et al., 2018	SC 4.0 digital transformations disruptions and strategy	Developed a theoretical framework to evaluate critical technologies and their impact on manufacturing companies' SC.
31	Isaksson et al., 2018	Digitalization and its effect on the future of control and operations	Addressed IoT's opportunities, challenges, and changes.
32	Frazzon et al., 2019	Towards SCM 4.0	Presented concepts and technological elements of SCM 4.0.
33	Majeed et al., 2017	IoT embedded future SC for I 4.0- An assessment from an ERP-based industry	Developed a conceptual framework to enhance inbound and outbound operations in ERP for the fashion and footwear apparel industry.
34	Scuotto et al., 2017	A buyer-supplier relationship in the context of SME's DSC management	Analyzed ICT-specialized Human Resource Management, knowledge-sharing activities, buyer-supplier relationship and adoption of the electronics market.
35	Ivanov et al., 2018	A survey of control theory applications to operational systems, SCM and I 4.0	Application of control theory to engineering and managerial problems in SC operations.
36	Cenamor et al.,2017	Adopting a platform approach in servitization leveraging the value of digitalization	Platform approach for advanced service operations in manufacturing firms.
37	Panitto et al., 2019	Cyber-physical manufacturing enterprises of the future	Listed future industry challenges- Development of new scientific control for SC network control.
38	Gupta et al., 2020	Dynamic Capabilities and Institutional theories for I 4.0	Explored the firm's orientation in adopting I 4.0 and DSC.
39	Tortorella et al., 2020.	Organizational learning paths based upon industry 4.0 adoption: An empirical study with Brazilian manufacturers.	They studied the mediating role of learning capabilities in an organization at different contextualization levels between digital technologies and operational performance.
40	Gustafsson, Jonsson, & Holmstrom, 2019	Digital Product Fitting in Retail SCs- Maturity levels and potential outcomes	Developed the maturity model of product fitting specifying three levels of digitalization and possible outcomes.
41	Ghobakhloo, 2020	I4.0, Digitization and Opportunities for Sustainability	Reviewed state-of-the-art technology trends of I 4.0 and introduced the architectural design of I 4.0.

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Sl No	Author	Aim/ Objectives	Implications and Contributions
42	Bechtsis et al., 2018	Intelligent autonomous vehicles in DSC- a framework	Developed a framework for the effective integration of intelligent autonomous vehicles.
43	Xue et al., 2013	Risk mitigation in SCD- System modularity and information technology governance	System modularity mitigates the risk of adopting a digital SC.
44	(Parida, Sjödin, & Reim, 2019)	Reviewing the literature on Digitalization, Business Model Innovation	Profiting from digitalization requires a business model; Developed an innovations framework.
45	Alcacer and Cruz-Machado, 2019	Scanning the I 4.0- Literature review on technologies for manufacturing systems	Scanning of I 4.0 environment - Described enabling technologies and systems over the manufacturing environment.
46	Buchi et al., 2020	Smart Factory Performance and I 4.0	Investment in industry 4.0 is in terms of high cost, acquisition of new skills and risk of obsolescence.
47	Brinch et al., 2021	Firm-level capabilities toward bid data value creation	Identified various capabilities related to process, performance, human, strategic, and organizational practices.
48	Min et al., 2019	Machine learning-based digital twin framework for production optimization	Proposed a theoretical framework for digital twin-based production optimization by integrating industrial IoT processing and machine learning.
49	Kittipanya-Ngam and Tan, 2020	A framework for food SCD	Explored the practices, challenges, and opportunities faced by Thailand food manufacturers in digitalizing their food SCs.
50	Tang and Veelenturf, 2019.	The strategic role of logistics in the industry 4.0 era.	Cyber-physical systems can be developed by leveraging digital technologies to gain a competitive advantage.
51	Seyedghorban et al., 2020	SCD: Past, present and future	Future research plans in SCD include data science-enabled SCM, SC agility, humanizing manufacturing through digital manufacturing strategy, Omni-channel and Internet of Things, and resource-based view and beyond.
52	Wilkesmann and Wilkesmann, 2018	Industry 4.0 – organizing routines or innovations?	Various applications of industry 4.0 are based on dimensions of the degree of formalisation, location of control authority, knowledge and degree of professionalism.
53	Zangiacomi et al. (2020	Investigated the managerial practices for the adoption of I 4.0 technologies	Addressed the key challenges, common mistakes and best practices according to the level of digital implementation.
54	Sanders et al., 2019	Sustainable SCs in the age of AI and digitization: research challenges and opportunities	Presented the current dominant themes in research on sustainable SCs in the age of digitization and potential new research opportunities.
55	Wilding and Wagner, 2014	Special Issue: building theory in SC management through systematic reviews of the literature	A systematic literature review facilitates evidence-based decision-making, focusing on SCD.
56	Zhao et al., 2019	Sustainability and digitalization of corporate management based on augmented/virtual reality tools usage: China and other world IT companies' experience	Analyzed the impact of digitalization on sustainability and corporate management based on augmented/virtual reality tools.
57	Ardito et al., 2019.	Towards Industry 4.0.	Detailed the novel efforts to develop digital technologies for managing the interface between SCM and various functions like sustainability, marketing process and integration.
58	Boehmer et al., 2020	The impact of the Internet of Things (IoT) on servitization: an exploration of changing supply relationships	Examined the potential of IoT in utilizing the potential for production, business and operating models.

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Sl No	Author	Aim/ Objectives	Implications and Contributions
59	Attaran, 2017	The Rise of 3-D Printing: The Advantages of Additive Manufacturing over Traditional Manufacturing.	Identified additive manufacturing implementation challenges, technologies and trends; discussed advantages of additive manufacturing over traditional manufacturing.
60	Bordeleau et al. 2020	Business intelligence and analytics value creation in Industry 4.0: a multiple case study in manufacturing medium enterprises.	Business performance in manufacturing medium enterprises can be leveraged through business intelligence and analytics.
61	Cavalcante et al., 2019	A supervised machine learning approach to data-driven simulation of resilient supplier selection in digital manufacturing.	Proposed the role of a hybrid approach of machine learning and simulation in creating digital twins and its applications to data-driven decision support systems in resilient supplier selection.
62	Chen et al., 2019	Impact of SC information sharing on the performance of fashion enterprises.	Proposed a novel research model explaining the relationship between information sharing, SC integration, business performance and operational performance.
63	Chen and Lin, 2017.	Feasibility evaluation and optimization of a smart manufacturing system based on 3D printing: A review.	Analyzed smart manufacturing systems based on 3D printing; found that the technical issues should be addressed to safeguard the viability of 3D printing applications in the manufacturing context.
64	Di Vaio and Varriale, 2020	Digitalization in the sea-land SC: experiences from Italy in rethinking the port operations within inter-organizational relationships.	Highlighted the implications of digital platforms in redesigning and rethinking various processes in business considering the inter-organizational relationship among SC partners.
65	Dolgui et al., 2019	Scheduling in production, SC and Industry 4.0 systems by optimal control: fundamentals, state-of-the-art and applications.	Presented the applications of optimal control in production scheduling, Industry 4.0 and SC by focusing on the deterministic maximum principle.
66	Dunke and Nickel, 2020	Improving company-wide logistics through collaborative track and trace IT services.	Explored the application of collaborative information technology applications in enhancing intra-organizational logistics.
67	Frank et al., 2019	Industry 4.0 technologies: Implementation patterns in manufacturing companies	Explained the pattern of adoption of digital technologies in manufacturing firms considering the base and front-end technologies.
68	Hastig & Sodhi, 2020	Blockchain for SC traceability: Business requirements and critical success factors	Presented a measurement model for practical work and a base for future research on blockchain applications for SC traceability.
69	Heinen and Hoberg, 2019	Assessing the potential of additive manufacturing for the provision of spare parts.	Followed a universal perspective approach in developing a design to analyze the impact of additive manufacturing.
70	Hofmann and Rusch, 2017	"Industry 4.0 and the current status as well as prospects on logistics	Analyzed the opportunities for logistics in the context of Industry 4.0 and its implications.
71	Ishtiaque et al., 2020	'Impact of the technology-based integrated, responsive SC on operational performance: a case of a volatile market	Developed a comprehensive model that aids in establishing and understanding the ICT, operational responsiveness, integrative capabilities and performance dimensions in the manufacturing industry.
72	Joshi and Gupta, 2019	Evaluation of design alternatives of End-Of-Life products using the internet of things.	Developed an advanced production system using IoT, which satisfies various components, products and materials demands.

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Sl No	Author	Aim/ Objectives	Implications and Contributions
73	Kache and Seuring, 2017	Challenges and opportunities of digital information at the intersection of Big Data analytics and SC management	Investigated the potential impacts of Big Data Analytics on information usage in a corporate and SC context.
74	Kamble et al., 2020	Big data-driven SC performance measurement system: a review and framework for implementation	They found that performance measures and metrics can enhance the firm's performance based on effective communication among various SC partners in the network.
75	Kamble et al., 2018.	Sustainable Industry 4.0 framework: A systematic literature review identifying the current trends and future	They gave the current status of research in the domains of Industry 4.0.
76	Zutshi and Grilo; 2019	The emergence of Digital Platforms: A Conceptual Platform Architecture and impact on Industrial Engineering	Their research conceptualized the digital platform and its effect on industrial engineering and digital platforms' key aspects, characteristics, and value proposition.
77	Liao et al., 2017	Past, present and future of Industry 4.0 a systematic literature review and research agenda proposal.	They summarized the current research activities indicating deficiencies and the potential research directions in Industry 4.0.
78	Luz Martin-Pena et al., 2018	The digitalization and servitization of manufacturing: A review on digital business models	The digitalization and servitization of manufacturing lead to competitive advantages through innovative digital business models.
79	Mendhurwar and Mishra, 2021	Integration of Social and IoT Technologies: Architectural Framework for Digital Transformation and Cyber Security Challenges	Analyzed the interplay and synergetic relationships among digital technologies; identified relevant interactions and key challenges, primarily focusing on cybersecurity and privacy.
80	Pan et al., 2020a	Blockchain technology and enterprise operational capabilities: An empirical test.	Analyzed the relationship between blockchain technology and enterprise operational capabilities based on actual data.
81	Queiroz et al., 2019	Industry 4.0 and digital SC capabilities.	Identified the basic capabilities and enablers that shape the digital SC capability framework.
82	Queiroz et al., 2020	Smart production systems drivers for business process management improvement: An integrative framework	Studied the key drivers from the literature associated with SPSs that impact the improved BPM.
83	Sarkis, Kouhizadeh, & Zhu, 2020	Digitalization and the greening of SCs. Industrial Management & Data Systems	Highlighted a reflective overview of traditional and emerging digitalization's role in leveraging environmental SC sustainability.
84	Sundarakani et al., 2019.	Designing a hybrid cloud for SC network of Industry 4.0: a theoretical framework.	Highlighted the significance of cloud computing in industry 4.0; Developed a SC cloud model for transforming SC partners towards Industry 4.0.
85	Malik, Tripathi, Kar, & Gupta, 2021	Artificial intelligence and its impact on employees working in Industry 4.0	Developed a practical understanding of the employee experiences due to AI adoption and technostress.
86	Sarkar et al., 2022	Port logistic issues and challenges in I 4.0 for emerging economies	Studied the issues faced by various stakeholders in port logistics; Developed a conceptual framework for managing port logistics issues.
87	Nica et al., 2021	Evaluate and analyze IoT sensing networks, AI-based decision making algorithms	Analyzed and found the impact of IoT and AI-based decision making algorithms and real-time process monitoring in sustainable I 4.0.

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Sl No	Author	Aim/ Objectives	Implications and Contributions
88	Mir et al., 2020	Identify critical success factors in developing intelligent autonomous systems	Developed a model for prioritization hierarchy for building a sustainable ecosystem for intelligent autonomous systems through AI.
89	Lee and Lim., 2021	To analyze and understand I 4.0 through text mining with unsupervised machine learning algorithms	Identified research and application issues related to I 4.0, categorized within a five-level hierarchy
90	Novak et al., 2021	Explore product decision making information systems in sustainable I 4.0	Developed product information systems, real-time sensor networks and artificial intelligence-driven big data analytics in sustainable I 4.0.
91	Becue et al., 2021	Identify opportunities and threats of using AI in the manufacturing sector	Detailed essential drivers and principles of I 4.0 in the advent of AI in manufacturing systems.
92	Kar et al., 2022	To comprehensively review AI and sustainability	Provided information on the industry-specific application of AI for sustainability.
93	Grover, Kar, & Dwivedi, 2020	Understanding AI adoption in operations management	Explored the feasibility of AI utilization in operations management elements like manufacturing, product development, services and supply chain.
94	Deepu and Ravi, 2021	Developing a model for the selection of IOIS for SCD	Developed an Integrated model for the selection of IOIS for SCD.
95	Kumar, Vrat, & Shankar, 2021	Prioritization of strategies for overcoming I 4.0 barriers	Proposed a hybrid framework for overcoming the barriers in I 4.0.
96	Sarkar and Shankar., 2021	Analyze barriers associated with port logistics in I 4.0	Developed a hierarchical model that reveals the inter-dependencies among the barriers for effective operation in I 4.0 and data-driven decision making.
97	Konstantakopoulos et al., 2021	Analyzing sustainable approaches for logistics companies	Contributed towards identifying and developing strategies to enhance efficiency and accuracy in achieving green logistics
98	Kushwaha et al., 2021	Review on application of big data in emerging management disciplines	Uncovered the emerging management areas supported by big data.
99	Han et al., 2021	Identify and focuses on the challenges and solutions in digitizing supply chains	The solutions developed were assessed based on data obtained from automobile manufacturers in China. The results of the study reveal that these solutions can make a positive and significant impact on SCD.
100	Wamba, S. F., & Queiroz, 2022	Determine the blockchain diffusion across supply chains.	Explored the essential differences in the variables that determine blockchain innovation across countries.
101	Kunkel et al., 2022	Digitalization and its implications for sustainable supply chain collaboration.	They found that a wide range of digital technologies at various maturity levels is used in sustainable supply chain collaboration, which needs to be prioritized for addressing sustainability.
102	Alsufyani & Gill, 2022.	Impact of digitalization on organization performance outcomes	Identified performance indicators, measures, metrics and scales that impact digitalization.

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